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NUMERICAL AND DEVELOPMENT APPENDIX TO NEURAL BELLMAN OPERATORS

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This appendix retains the economic-inference, safeguarded-selection, localization, certified-readout, deployment, recursive-query and vector-control sections without textual deletion. It records their original scientific evidence and independent-verification boundaries. New exact action-switch duals and first-disagreement proofs establish the optimizer-class cost transfer. The complete prospective factorial catalogue reports all ninety-four original-law cost estimands, precision components and construction-work boundaries. Core proofs remain in the Technical Supplement; the original broader applications remain in their unchanged companion editions.

APPENDIX A: FROM RETURNED POLICIES TO ECONOMIC RESOURCE DECISIONS

An all-state upper bound and a comparison of actual policy costs answer different economic questions. We now connect the policies returned by a construction frontier to direct cost inference and then to a choice of implementation resources. The construction data select the policies; independently generated evaluation paths assess them. This separation permits a valid comparison of first-crossing policies with different resolutions.

A.1. *Simultaneous inference after construction*

Let $\mathcal{H}$ denote all construction information: primitive inputs, numerical labels, fitted continuations, returned policies and complete certificate ladders. Conditional on $\mathcal{H}$, fix a finite catalogue of implemented-policy contrasts. Each implementation includes its observation rule, original witness or nearest-node selector, true-state feasible repair and action quantization. For a contrast i, let D_i be the difference in

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expected discounted costs under a specified initial law. A common-innovation path score evaluates the two policies on their own endogenous trajectories. It is not the difference between two regret bounds.

Suppose the initial and innovation uniforms are partitioned into equal bins. Independent uniform bin indices are drawn, and interval simulation encloses the score for every continuous realization inside the drawn bins. All unresolved state, selector and repair branches remain in the enclosure. If $[A_i, B_i]$ is a deterministic support interval, clip the path endpoints to that support. Write $\underline{m}_i, \overline{m}_i$ for outward sample-mean bounds on the lower and upper endpoint samples and $v_{i,L}, v_{i,U}$ for upper bounds on their unbiased sample variances. For $n \geq 2$ define

$$R_i(v) = \sqrt{2v\ell/n} + \frac{7(B_i - A_i)\ell}{3(n-1)}, \qquad \ell \geq \log(4K/\alpha), \tag{1}$$

where K bounds the number of sampled contrasts.

THEOREM 1—Construction-conditional economic intervals: *Under the stated independent-bin model, conditional on any valid construction history $\mathcal{H}$, with probability at least $1 - \alpha$ all catalogue contrasts satisfy*

$$D_i \in [l_i, u_i] := [\max\{A_i, \underline{m}_i - R_i(v_{i,L})\}, \min\{B_i, \overline{m}_i + R_i(v_{i,U})\}]. \tag{2}$$

The same coverage holds unconditionally. Subsequent selection among these covered contrasts, or evaluation at any deterministic implementation-price vector, needs no additional union bound. This conclusion applies only to policies and initial laws included in the catalogue; it does not create direct-cost observations for every unstudied tolerance or initial state.

The proof is in Supplement, Section 2.10.

In particular, selecting the first policy to meet a specified certificate is permitted when that selection uses construction records rather than the subsequent evaluation paths. Repeated construction clocks are not independent draws from a neural-optimizer population. The statistical randomness here belongs to the evaluation design. A fixed pseudorandom stream implements a reproducible experiment; it is not itself a proof of the independent-bin assumption.

A.2. *Replacement-fee identification*

Write $D = J^W - J^F$. If witness is installed, replacing it by FVI changes total cost by $k - D$, where $k \geq 0$ is the replacement fee. If FVI is installed, the change is $k + D$. Thus the two directional gains before fees are D and $-D$.

COROLLARY 2—The break-even fee frontier: *On an event where $D \in [l, u]$, the two directional gains belong to $[l, u]$ and $[-u, -l]$. The unknown absolute cost difference satisfies*

$$|D| \in [\text{dist}(0, [l, u]), \max\{|l|, |u|\}]. \tag{3}$$

For every $k \geq \max\{0, u, -l\}$ neither replacement strictly lowers expected cost. By contrast, $k \geq \max\{0, l, -u\}$ only removes a certified *strictly profitable replacement. The two thresholds generally differ. Strict inequalities in the first criterion certify that both replacements strictly increase cost.*

The proof is in Supplement, Section 2.11.

The fee frontier is more informative than a conclusion at one generous charge. The earlier $1/64$ replacement decision is retained with its original interpretation and confidence account. The new analysis reports the two directional gain intervals and both thresholds instead of selecting a favorable fee after seeing the data. There is no welfare calibration hidden in the units: a decision maker must supply the economically relevant charge, or use the full frontier to see what remains unidentified.

A.3. *Observation costs on the same bit grid*

Let $\mathcal{B} = \{4, \ldots, 16\}$ be the available bit precisions. For one fixed returned controller, write J_b for the expected cost of its actual b-bit, robustly repaired and quantized implementation. Use $b_0 = 16$ as a common reference and obtain simultaneous intervals for $D_b = J_b - J_{b_0}$. The identity interval at $b = b_0$ is exactly $[0, 0]$. Let $s_b = db \sum_{t<T} B_t$ be the discounted number of coordinate-bits purchased; the net cost is $J_b + \lambda s_b$.

THEOREM 3—Certified choice of actual net implementation cost: *Suppose all* $D_b \in [l_b, u_b]$ *simultaneously. For any* $\lambda \geq 0$ *define*

$$L(\lambda) = \min_b \{l_b + \lambda s_b\}, \qquad U(\lambda) = \min_b \{u_b + \lambda s_b\}, \tag{4}$$

$$\widehat{b}(\lambda) = \min \arg\min_b \{u_b + \lambda s_b\}.$$

Then, simultaneously for all $\lambda \geq 0$,

$$0 \leq J_{\widehat{b}} + \lambda s_{\widehat{b}} - \min_{b \in \mathcal{B}} (J_b + \lambda s_b) \leq U(\lambda) - L(\lambda). \tag{5}$$

The lower and upper functions are finite piecewise-affine envelopes. All changes in their active lines occur at nonnegative intersections of catalogue lines, with exact comparisons deciding ties. The smallest selected precision $\widehat{b}(\lambda)$ *is weakly decreasing in* λ.

The proof is in Supplement, Section 2.12.

This result makes a statement about *actual* net cost within the declared bit catalogue. It differs from minimizing a uniform all-state loss bound. Both criteria are useful: the latter gives distribution-free protection relative to the full-information optimum, while (5) uses a specified initial law to compare feasible implementations of one returned policy. Neither criterion gives an optimum over arbitrary sensor technologies or all policies.

For a fair comparison the conventional actor receives its own acquisition allowance. Suppose its nearest-node policy has a certified selected-policy account at a reported state, obtained from the graph modulus L_t^{gr}, and its actual interpolated critic has modulus L_t^f. Transporting the same selected node action to the true state, then using cell-wide capacity repair, adds at most

$$(L_t^{\mathrm{gr}} + L_t^f + 2D_t K_t^A) r(b) + D_t \Delta_t. \tag{6}$$

Indeed, transport the feasible node pair directly to the true state, bound its node distance by the reported-state distance plus $r(b)$, and bound $f_t(\widehat{x}) - f_t(x)$ by $L_t^f r(b)$. Robust repair adds $D_t(2K_t^A r(b) + \Delta_t)$. This proves (6) without assuming actor continuity. For a witness critic $L_t^f = L_t^{\mathrm{gr}}$, it reduces to the preceding witness allowance. The empirical comparison therefore evaluates the all-state-account minimizer, the

interval upper-net-cost selector, and a descriptive midpoint selector on the *same* integer bit and price grid. The midpoint selector is not called the true optimum.

APPENDIX B: CERTIFICATE-GATED NEURAL QUERIES

A prediction of the optimal action and a prediction of a useful numerical query are different objects. Inserting a learned action before examining an already available certificate charges an extra continuation query even when the two boundary actions suffice. We instead make the demand for another query a mathematical decision and let a trained NBO selector determine its location only after that demand has been established. The underlying continuation recursion and the economic optimum are unchanged.

At a recursive state, start with the two feasible action endpoints. Compute the global lower bound from the parabolic minorants of Lemma 13, including the capacity remainder, and the smallest upper endpoint. Return immediately when their difference is at most the requested tolerance. Otherwise choose a partition interval with smallest current minorant. A learned action is used only if it lies in the central half of that interval; it then replaces the conventional proposed split. Outside that region the safeguarded parabolic split is used. The rounded split must pass the one-fifth interior check before evaluation. A prediction is evaluated at most once at each recursive state and is reused only as a proposed location. It supplies neither a value endpoint nor a lower bound.

THEOREM 4—Gated prediction and complete-policy validity: *Under the assumptions of Theorem 14, the gated service returns the same original-optimum and actual-policy guarantees for every supplied prediction, independently of fitting error. In real arithmetic, with endpoint-plus-capacity width at most* $5\delta/8$*, a sufficient number of endpoint evaluations at a state is*

$$A_r(\delta, x) = 4 + \left\lceil 5c(x)\sqrt{H_r/(3\delta)} \right\rceil .$$

Each requested evaluation either reduces the unresolved action partition or terminates the query; no initial interior evaluation is mandatory. If the boundary certificate suffices, no predictor is evaluated at that state. If every prediction is declined at every recursive

query, the value endpoints and implemented actions coincide with those of the conventional parabolic service using the same arithmetic and tie rules.

The proof is in Supplement, Section 2.8.

This theorem establishes robustness to the numerical content of a fitted prediction, not that learning is free or always useful. It also does not state that an arbitrary routed transcript has fewer queries than the adaptive conventional transcript. The executed comparison records both possibilities. Unlike a policy catalogue with a stored reference action at every state, neither service constructs a conventional state lattice before it can return a certified policy.

B.1. *State dimension, working memory, and reliability*

For the original scalar model, the primitive bounds $dG_r \le 27$ and $dM_r \le 54$ imply

$$H_r \le \frac{3369}{256}, \qquad \Lambda_r \le \frac{1319}{128}, \qquad \frac{\beta M_{r-1} d}{6144} \le \frac{135}{16384}.$$

Thus the action and scalar-innovation budgets at fixed remaining horizon and tolerance are independent of the number of state coordinates. Every primitive evaluation still requires $O(d)$ arithmetic. A width-w one-hidden-layer ReLU prediction on the recorded $d+5$ features requires $O(dw)$ work and stored parameters; a full quadratic prediction on these features uses $O(d^2)$ terms. Neither cost is included for free in the Bellman recursion.

Let $q_r(\delta)$ be the dyadic quadrature count and let $P_r(d)$ be the cost of one predictor evaluation. Apart from the separately identified exact-arithmetic recovery, a conservative work recursion is

$$W_r(d,\delta) \le CdA_r(\delta) + P_r(d) + A_r(\delta) q_r(\delta) W_{r-1}(d, \delta/(4\beta)).$$

The predictor term is absent on a boundary-certified query. This expression removes the exponential state lattice, not the horizon or innovation-tree dependence. A depth-first scalar implementation can retain the current action partition and one innovation child at a time. The executed batched implementation instead reports its maximum state batch and process peak memory. A zero count of stored state-lattice nodes does not mean zero storage.

The deterministic certificate is valid for any supplied finite network weights. A failure to fit and return such weights is a separate service event. Our reliability estimand is the return count and worst measured complete work over the preregistered finite seed catalogue, with every repetition retained. Numerical workload paths are not optimizer replications. The real-arithmetic cap and the exact-rational precision condition establish algorithmic termination at admissible tolerances; recorded processor seconds are descriptive, not a hardware-uniform worst-case bound.

B.2. *An explicit economic purchase criterion*

The decision maker buys a callable controller rather than a table of successful sample decisions. Fix a physical capital numeraire for the normalized state, a conversion $m > 0$ dollars per unit of the model's loss index, a computation charge $p \geq 0$ dollars per measured second, and a fixed deployment fee $f \geq 0$ dollars. For n uses under a specified initial-state law, the monetary objective is

$$\mathcal{C}(\pi) = nm\,\mathbb{E}[J_0^\pi(X_0)] + pW(\pi) + f.$$

These are declared preference and service-price parameters, not estimated economic primitives. The policy theorem bounds the expected loss under the original continuous law for every such initial-state law. The finite dyadic workload below measures execution; its average cost is not substituted for that expectation.

If two policies have an independently valid expected-gain interval $[l, u]$ for $\mathbb{E}[J_0^{\pi_0} - J_0^{\pi_1}]$, and their incremental measured service charge is $\Delta = p(W_1 - W_0) + f_1 - f_0$, then the monetary net gain lies in

$$[nml - \Delta,\ nmu - \Delta].$$

This follows by multiplication by the positive numeraire conversion and subtraction of the same service charge. A positive lower endpoint supports purchase; a negative upper endpoint rules it out at the stated prices. Uniform regret certificates alone do not supply a directional expected-gain interval. They instead support a worst-case accuracy requirement imposed before selecting the least expensive eligible service.

Finally, a forecast of amortized query work is distinguishable from a measured complete service. With a fixed model and exactly repeated query workload, if training requires B operations and deployment requires D_L rather than D_C operations per workload, then learning uses fewer total operations after k repetitions precisely when $D_L < D_C$ and $k > B/(D_C - D_L)$. If $D_L \geq D_C$ and $B \geq 0$, no repetition of that unchanged workload produces an operation saving. This arithmetic criterion includes training and cannot be applied to unrelated future states without a new assumption. The executed catalogue reports its own full process clocks, not a favorable hypothetical amortization.

APPENDIX C: WHEN A PREDICTED QUERY COMPLETES THE CERTIFICATE

The safeguarded theorem is insensitive to the quality of a supplied prediction. A separate question is when that prediction is sufficient to finish verification. The answer must concern both the selected action and the unresolved lower bound. Proximity to a minimizer alone does not make a coarse lower certificate accurate.

PROPOSITION 5—Witness accuracy and certificate localization: *Let f be a differentiable, H-semiconcave objective with $H \geq 0$ on $[0, c]$, let a^* minimize f, and suppose $|f'| \leq \Lambda$. An evaluated proposal $\widehat{a}$ lies in $[0, \underline{c}]$, where $0 < \underline{c} \leq c$ and $c - \underline{c} \leq \Delta$. Each endpoint interval is a valid enclosure of f and has width at most w. Suppose the partition containing $\widehat{a}$ covers $[0, \underline{c}]$ and has maximum interval length h. Let L be its parabolic lower certificate with the capacity allowance $\Lambda\Delta$, and let U be the smallest upper endpoint, retaining earlier valid bounds as in the query algorithm. With $e = |\widehat{a} - a^*|$ and $\lambda = |f'(a^*)|$,*

$$U - L \leq 2w + \Lambda\Delta + \frac{Hh^2}{8} + \lambda e + \frac{He^2}{2}. \tag{7}$$

For an interior minimizer $\lambda = 0$. At a boundary the linear term cannot generally be removed. Thus the right side at most δ is a sufficient certificate-return condition, not an assumption that a trained prediction meets it.

If old and new valid endpoint records give (L, U) and (L', U') after one additional query, with the running maximum lower bound and minimum upper bound, then

$$(U - L) - (U' - L') = (U - U') + (L' - L) \geq 0. \tag{8}$$

An added prediction may have zero certified progress. The same identities apply to a conventional query.

The proof is in Supplement, Section 2.9.

This proposition isolates what a fitted representation can contribute without crediting it with a conventional solution. It can provide a small-error witness and change which unresolved interval is refined. The global partition, endpoint arithmetic and capacity account still have to support that witness. The maximum-width sufficient condition is deliberately conservative; the implemented minimum-minorant rule can stop earlier when only a subset of the partition remains competitive.

C.1. *A complete accuracy contract and its resource dependence*

For T dates and target ϵ, write $S_T = \sum_{t<T} \beta^t$. The recorded service chooses a common local tolerance by rounding $\epsilon/(2S_T)$ to binary64 and verifies the final outward allowance. Under $dG_r \leq 27$, a uniform per-date acquisition allowance at forty bits is

$$\alpha_b = \left(8 + \frac{297\beta}{16} + 27\right) 2^{-40}.$$

The service certifies $(\delta + \alpha_b)S_T \leq \epsilon$. Each descendant request is $\delta/(4\beta)$, rounded as in the actual code and included in its outward endpoint guard. At the smallest request in the two- and three-date catalogues, the declared thirty-two-bit action quantum satisfies the exact-recovery condition $\delta_r > 8\Lambda_r 2^{-32}$. The supplement checks this inequality with rationals for every horizon and target, rather than inferring it from a successful workload.

For fixed remaining horizon and accuracy, the bounds on action endpoints and scalar-innovation bins are independent of state dimension in this normalized economy. The factors dG_r and dM_r are bounded, while every transition and cost still reads d coordinates. A width-w_θ one-layer ReLU predictor has $O(dw_\theta)$ parameters

and evaluation work; a dense quadratic on the same features has $O(d^2)$ terms. These statements identify the absence of a $(n+1)^d$ stored state table. They do not count an arbitrary-precision rational operation as a unit-cost floating operation and do not suppress the product of descendant query counts as horizon grows.

The two fitted implementations and their conventional controls therefore share one original-optimum contract, but need not share a transcript, a returned policy or a runtime. The completed experiments charge their distinct transcripts. A finite-catalogue reliability statement concerns all specified seeds and repetitions, including fitting warnings. It is not a population statement about optimizer success.

APPENDIX D: MATCHED READOUTS AND COMPLETE RECOVERY

For $x \in [0,1]^d$, the two fixed feature maps used in the comparison are

$$\phi^{\mathrm{R}}_{jh}(x) = \frac{1}{d}(1 - |4x_h - j|)_+, \qquad \phi^{\mathrm{B}}_{jh}(x) = \frac{1}{d}\binom{4}{j} x_h^j (1-x_h)^{4-j}, \quad 0 \le j \le 4,$$

with $1 \le h \le d$. Both are nonnegative partitions of unity with $5d$ box readout coefficients. The first is an additive hat basis with an exact ReLU realization; the second is a degree-four Bernstein basis. Both purchase the same verified boundary contexts and approximate action labels, and optimize the same certificate loss. The label-loss control changes the objective, not its hat features. The achieved rational optimization certificate is evaluated after quantization.

D.1. *Complete recovery under a validation law*

The validation design draws each coordinate as $U/2^{16}$, with U uniform on $\{0, \dots, 2^{16}-1\}$. These states are representable by the twenty-four-bit policy acquisition contract, but are not uniform over that finer grid. The following result applies to the declared independent context law and recovery. It does not assert that on-policy recursive states have that law.

PROPOSITION 6—A finite catalogue of complete work: *Condition on a frozen finite family of predictors and a predictor-independent validation law. Suppose a failed screen invokes a deterministic recovery whose root-query count is at most $K(z)$, and every*

root query costs at most $W(z)$ under the declared descendant and arithmetic contract. If $0 \le W(z)K(z) \le C_{\max}$, then with probability at least $1-\alpha$, simultaneously for M fixed predictors,

$$\mathbb{E}[\textit{recovery work}_\theta] \le \frac{1}{n}\sum_{i=1}^{n} W(z_i)K(z_i)\ell_\tau(z_i,\theta) + C_{\max}\sqrt{\frac{\log(M/\alpha)}{2n}}.$$

Training, context acquisition, prediction, loading, identity checks and output costs are additional terms in complete work. Replacing W by one gives a root-query result, not a bound on elapsed process time.

Proof. See Supplement, Section 4.1.

The executed validation uses the root-query specialization and records descendant queries separately. It does not fit a runtime model and declare it a valid worst-case endpoint cost. The predictable-deployment result in Section E adds an explicit context-transfer premise and a complete operation account.

D.2. *An additional vector witness*

In the two-control economy a trained, projected and downward-rounded proposal can supply a new evaluated upper witness. Its insertion leaves the triangular lower cover unchanged. The original corner endpoints remain in that cover.

PROPOSITION 7—Optional vector witnesses: *Suppose a triangular action cover has a valid global lower bound L and feasible evaluated witnesses with upper bound U. Adding any finite number of feasible, correctly evaluated witnesses and replacing U by the minimum of all their upper bounds preserves containment of the original feasible infimum. Every returned witness with $U - L \le \delta$ has original Bellman regret at most δ. The conclusion is independent of the training method and of whether an added witness is a vertex of the lower cover.*

Proof. See Supplement, Section 4.2. The vector benchmark changes the trained proposal or the choice of a triangle to refine, not the authority of the global lower cover. Its training objective remains action-label loss.

APPENDIX E: FROM VALIDATED READOUTS TO COMPLETE DEPLOYMENT

The fixed-feature construction and its optimization certificate identify an achieved numerical object. The next issue is its use after training. Independent validation need not imply that the states subsequently visited by a controller are independent. We give a work account that keeps those two sources of information separate and charges the descendants of a failed screen.

E.1. *Predictable selection and changing context laws*

Condition on a frozen family Θ of M readouts. Write

$$f_\theta(z) = K(z)\ell_\tau(z,\theta), \qquad 0 \le N_\theta(z) \le f_\theta(z) \le K_{\max},$$

where the fallback and its endpoint cap are the ones specified for the context. Let $Z_1^v, \dots, Z_n^v$ be independent validation contexts with law P, independent of readout construction. Set

$$u_\theta = \min\left\{ K_{\max},\ \frac{1}{n}\sum_{i=1}^{n} f_\theta(Z_i^v) + K_{\max}\sqrt{\frac{\log(M/\alpha)}{2n}} \right\}. \tag{9}$$

During deployment, let $\mathcal{F}_{b-1}$ contain construction, validation, and all observations before block b. The selected readout θ_b is $\mathcal{F}_{b-1}$-measurable. The new context Z_b can depend on this history. Suppose a declared predictable number $\rho_b \in [0,1]$ satisfies

$$\|\mathcal{L}(Z_b \mid \mathcal{F}_{b-1}) - P\|_{\mathrm{TV}} \le \rho_b, \qquad \|P - Q\|_{\mathrm{TV}} = \sup_A |P(A) - Q(A)|. \tag{10}$$

This is a distributional condition, not a consequence of matching a model hash.

Use an additive operation model. Let F_0 bound fitting, serialization and validation work. Let h_b bound loading, identity checks, prediction, previously purchased endpoints, acquisition and durable output at block b. Finally let a nonrandom c_b bound the complete descendant and numerical-recovery work of one additional root endpoint. The budgets must cover all admissible histories, not merely the largest observed runtime. Assume the deployed work satisfies

$$W_B \le F_0 + \sum_{b=1}^{B} \{h_b + c_b N_{\theta_b}(Z_b)\}.$$

This includes a complete-action mesh fallback when that is the fallback validated by the experiment. Substitution of a different fallback requires its own cap and loss contract.

THEOREM 8—Validated work under predictable deployment: *For a fixed deployment window B and $0 < \alpha, \gamma < 1$, with probability at least $1 - \alpha - \gamma$,*

$$W_B \le F_0 + \sum_{b=1}^{B} h_b + \sum_{b=1}^{B} c_b \min\{K_{\max}, u_{\theta_b} + K_{\max}\rho_b\} + K_{\max}\sqrt{\frac{1}{2}\log(1/\gamma)\sum_{b=1}^{B} c_b^2}. \tag{11}$$

The readout can be selected using the validation sample and the preceding deployment history. The conclusion does not require the deployment contexts to be independent. Under the same local feasibility and certificate premises, this selection does not change the original-optimum policy guarantee.

Proof. See Supplement, Section 3.7.

This theorem gives a prospective route from training to complete work rather than interpreting a training loss as a timing result. A smaller root-query mean is useful when its descendant-weighted savings exceed setup and per-block overhead. The c_b terms expose the continuation-tree cost. Setting them to one would prove a root-query statement, not a full-service statement. The R69 independent-context experiment below uses its declared mesh recovery; its adaptive recursive deployment records are not claimed to satisfy an empirically verified version of (10).

E.2. *Policy identity, declined reuse, and adoption*

Cost inference can be combined with a deployment clock only after identifying the controller whose cost was estimated. The release checks the complete readout arrays, tolerance, acquisition precision, action rounding and original kernels. Consider a service processing B blocks of m trajectories each. In D predetermined blocks it declines an incompatible readout and uses the benchmark ν; otherwise it uses a fixed π. Suppose each block has the same initial law and economic primitives, and

the decline schedule is independent of the trajectories. Let $[l, u]$ be a simultaneous interval for $\Delta = \mathbb{E}[J^{\pi} - J^{\nu}]$.

PROPOSITION 9—Complete-service adoption region: *Let W_{π}, W_{ν} be complete work observations under the same boundary. If $v \geq 0$ values one normalized expected-cost unit, $p \geq 0$ prices one work unit, and $f \geq 0$ is the incremental installation fee, then the net cost difference belongs to*

$$[\, vm(B - D)l + p(W_{\pi} - W_{\nu}) + f,\ vm(B - D)u + p(W_{\pi} - W_{\nu}) + f \,] \tag{12}$$

on the original simultaneous event. The statement holds for every such price triple on that event. With observed work saving $S = W_{\nu} - W_{\pi} > 0$, strict adoption is supported whenever

$$pS > f + vm(B - D)u.$$

If work is instead bounded by valid intervals, replacing $W_{\pi} - W_{\nu}$ by its interval difference preserves the statement.

Proof. See Supplement, Section 3.8.

The executed reuse design has $B = 64$, $m = 8$, and one predetermined identity rejection, so its normal-policy multiplier is 504. This does not make an observed prefix clock a completed service, or an observed timing range a confidence interval for future hardware. The paired complete receipts and the conservative envelope across the observed repetitions are both reported. The inputs v, p, f remain declared decision parameters; the experiment estimates neither an installation fee nor the monetary value of normalized investment losses. The policy-cost margin supplies an economically interpretable eligibility condition, while the full adoption region keeps computational savings and policy quality separate.

APPENDIX F: TRAINING, REUSE, AND POLICY-COST PRECISION

The experiment links the constructive readout to the actual controlled economy. It compares representation and training loss at a common information budget, validates a frozen readout on independent contexts, measures a longer deployment window, and evaluates the returned controllers at a prespecified policy-cost margin.

The original continuous laws, state-dependent investment capacity and Bellman target are unchanged.

F.1. *Design and information balance*

The scalar tasks are $(d, T, \varepsilon) = (2, 2, 1/128)$ and $(8, 3, 1/64)$, with twenty-four-bit state acquisition and thirty-two-bit downward action rounding. Four fitted proposals receive the same forty-eight training states, approximate action labels and verified boundary endpoints per nonterminal date. The certificate-aware hat–ReLU and degree-four Bernstein readouts have exactly $5d$ box-constrained coefficients and the same objective, penalty and optimizer budget. An action-label readout on the same hat features separates the objective from the representation. A quadratic label fit retains its explicitly different parameter count. Adaptive parabolic search and upper-chord minimization are prediction-free controls; unused training is not charged to them.

The hat basis has an exact finite ReLU realization. It also has a conventional piecewise-linear interpretation, so this comparison does not count two encodings of that basis as independent algorithms. The Bernstein comparison holds the coefficient count and purchased information fixed. All fitted proposals use the same final certificate. Every returned certificate-aware iterate is quantized and receives an exact rational value of the objective and box dual gap from Proposition 17. The achieved gap is reported rather than assuming that an optimizer's success flag proves optimality.

The protocol fixes two fitting seeds and four process repetitions. It contains eighty cold comparisons, sixteen independent-validation services, fifty-six reuse services, two paired-inference services and thirty-four vector services: 188 complete processes. All returned. This is finite-catalogue evidence for the complete verifier, not a population claim about fitting success. Four clock repetitions are not four new training samples. The construction, receipts and original source freeze predate this manuscript integration; the publication replay adds no scientific observations.

The complete cold-start results are reported in Supplement Table L.I. At a fixed purchased-information and coefficient budget, the hat–ReLU readouts close 48 and

50 of 256 small-task validation contexts for the two seeds, compared with 25 and 21 for Bernstein. The achieved exact box dual gaps are at most 10^{-7} across the certificate-aware catalogue. This is a method-specific improvement in the declared context screen, not a uniform complete-runtime advantage. All methods and both tasks remain in the supplementary validation table.

Cold-start work remains an exacting comparison. In the two-state task the median complete hat–ReLU process takes about 0.465 seconds, against 0.365 for adaptive search. In the eight-state task the respective medians are about 1.618 and 0.666 seconds; both use 2,594 policy-query evaluations in the displayed workload. These counts exclude separately reported training queries. Thus the certified objective is now an executed arm, but its execution is not an identified complete-work improvement. The recorded fitting, prediction and descendant counts locate the cost rather than attributing every successful return to a successful predictor.

F.2. *Prospective validation and reuse*

Each fitted readout is frozen before a new predictor-independent context stream drawn uniformly from the sixteen-bit dyadic state grid is generated. There are 256 contexts per nonterminal date. The validation calculation executes the declared complete-action mesh recovery whenever the pre-query screen fails. Its clipped loss bounds the additional root queries on every recorded context. The supplementary table gives the achieved loss, actual recovery queries, simultaneous validation upper bound and descendant counts for every task, date, method and seed. These 6,144 method–context–date evaluations validate the stated context law and fallback; they are not relabeled as independent on-policy recursive states.

Each reuse process trains and serializes once and then serves sixty-four predetermined blocks of eight trajectories. At block index 16 it deliberately receives a stale acquisition identifier and uses the adaptive benchmark. Loading, hashing, identity validation, rejection, queries, cash-path enclosures, durable files and process shutdown are included. A stable observed crossing must begin by block index 48 and remain below the matched adaptive prefix through index 63. This requirement is fixed before execution.

TABLE F.I

SIXTY-FOUR-BLOCK PROSPECTIVE REUSE

(d, T)	Rule	Policy queries	Median (s)	Range (s)	Wins	Stable
2,2	A	10057	3.621	[3.169, 7.129]	–	–
2,2	B	10392–10502	3.521	[3.320, 3.572]	4/8	4/8
2,2	H	10332–10447	3.746	[3.320, 5.075]	4/8	3/8
2,2	Q	9982–9992	3.621	[3.120, 4.473]	4/8	3/8
8,3	A	218656	24.914	[24.762, 25.216]	–	–
8,3	B	220282–220579	26.444	[26.068, 26.921]	0/8	0/8
8,3	H	220498–222019	26.469	[25.915, 27.568]	0/8	0/8
8,3	Q	223929–226193	27.872	[26.964, 28.973]	0/8	0/8

A complete-time win is relative to the adaptive process with the same repetition index. Stable means an observed cumulative crossing by index 48 that persists through index 63. The last summary and shutdown belong to complete clocks, not prefix clocks. The stale identity at index 16 is declined in every process. The same adaptive repetition is shared across matched seeds; these comparisons are not independent trials.

Three of the eight hat–ReLU processes in the two-state task satisfy that stable-prefix criterion; four have smaller final complete times than their matched adaptive process. The Bernstein and quadratic rules also exhibit small-task crossings. None of the learned processes wins the eight-state reuse comparison. In the small task, the adaptive complete times range from about 3.17 to 7.13 seconds, while hat–ReLU ranges from about 3.32 to 5.07 seconds. Consequently, a favorable matched crossing cannot be interpreted as uniform amortized superiority. The operation counts and every unfavorable repetition remain in the record.

The deployment theorem states what additional context-law and descendant-cost information would be sufficient for a prospective work bound. The experiment does not substitute its identity check for a distributional drift bound. Likewise, the sixty-four-block window is not extrapolated to an unobserved stopping horizon.

F.3. *A cost comparison at the declared precision*

The central inference change is the Bellman-centered observation of Theorem 1. It estimates the actual expected cost difference by evaluating optimal advantages along each actual policy trajectory. The conditional Bellman expectations are evaluated with a tighter tolerance and paid for separately. The estimator is not the difference between two policy-loss upper bounds, and its realized observation is not asserted equal to a same-shock cash-cost difference.

For each scalar task the protocol fixes 1,024 new common-path rows, forty-eight-bit continuous bins and evaluation tolerance $\varepsilon/32$. The policies are adaptive search, certificate-aware hat–ReLU, certificate-aware Bernstein, and the original R67 ReLU controller at its pinned fitting seed. That historical controller is loaded unchanged. The predeclared loss margin is $1/256$, with numerical and sampling half-width allocations of $1/512$ each. No new row is added after inspecting a sign or a failed comparison. There are twelve paired contrasts across the two tasks on one centered-inference event with failure probability at most 0.01.

All twelve centered intervals lie within $[-1/256, 1/256]$. They therefore establish two-sided equivalence at the declared normalized-loss margin, including comparisons involving the unchanged historical neural controller. Every interval still contains zero: strict cost superiority is not identified. Both precision allocations are met. This replaces the earlier inability to assess the declared margin by a positive, original-law eligibility result, without changing the economic estimand or assuming a fitted value is exact.

For adaptive-minus-hat–ReLU, the two-state interval is approximately $[-0.00068263, 0.00070764]$ and the eight-state interval is approximately $[-0.00160291, 0.00160253]$, with outward display rounding. The corresponding cash-cost intervals on the same new rows have widths about 0.423 and 0.529. The centered widths are about 0.00139 and 0.00321. This is an identified precision improvement for the same expected-cost target, not an identified neural speed advantage. The complete shared inference processes cost about 9.03 and 408.63 seconds, respectively. Conditional Bellman integration, rather than state-bin width, dominates their numerical enclosure budget.

TABLE F.II

BELLMAN-CENTERED EXPECTED POLICY-COST DIFFERENCES

(d, T)	Pair	Simultaneous interval	Numerical width	Radius	Margin
2,2	A–B	[-0.000683, 0.000709]	0.000194	0.000599	Yes
2,2	A–H	[-0.000683, 0.000708]	0.000194	0.000599	Yes
2,2	A–R	[-0.000658, 0.000728]	0.000188	0.000599	Yes
2,2	B–R	[-0.000670, 0.000715]	0.000187	0.000599	Yes
2,2	H–B	[-0.000695, 0.000695]	0.000193	0.000599	Yes
2,2	H–R	[-0.000670, 0.000715]	0.000187	0.000599	Yes
8,3	A–B	[-0.001603, 0.001603]	0.000812	0.001197	Yes
8,3	A–H	[-0.001603, 0.001603]	0.000812	0.001197	Yes
8,3	A–R	[-0.001583, 0.001613]	0.000801	0.001197	Yes
8,3	B–R	[-0.001583, 0.001613]	0.000802	0.001197	Yes
8,3	H–B	[-0.001603, 0.001603]	0.000812	0.001197	Yes
8,3	H–R	[-0.001583, 0.001613]	0.000802	0.001197	Yes

Pair first minus second; R is the unchanged R67 ReLU readout at seed 6603. Every task uses 1,024 new common-path rows, not that number per pair. All intervals are outward-rounded. Margin denotes containment in the prospectively specified interval $[-1/256, 1/256]$, not strict superiority. The numerical term is a full width; the sampling term is a half-width.

The two controllers' common initial-value terms cancel in expectation. All compatible true states within a sampled bin remain covered. No new inference trajectory in this catalogue has an unresolved acquisition branch. The raw conditional-value width, propagated bin width, acquisition allowance and final arithmetic diagnostics are reported separately; they are not assumed statistically independent, and pre-intersection component widths are not falsely presented as an additive decomposition of the tighter final interval.

F.4. *Two controls and the economic service*

The vector study compares simplicial search, largest-edge uniform refinement, a trained two-output hat readout and a trained two-output quadratic proposal at two state dimensions and horizons two and three. Both learned rules purchase the

same twelve teacher states per nonterminal date. Their training objective is action-label loss, not the scalar convex certificate loss. All receive the same triangular lower certificate and feasible rounding rule under two independent continuous innovations. The horizon-four hat proposal transfers its three-date model and is identified as a transfer, not as a four-date fitted model.

Supplement Table L.II reports every two-control workload.

The vector records contain actual continuous-bin policy paths, complete clocks, peak resident memory and descendant counts. Eight paths per ordinary service, and one per horizon-four transfer service, are workload diagnostics rather than a precision-designed population comparison. The experiment adds a trained vector comparison, but it does not identify a vector expected-cost ranking or remove continuation-tree growth. Derivative branch-and-bound is not an executed arm of this catalogue.

For the scalar deployment decision, the release verifies that the cost-inference readouts coincide with the relevant cold and reuse readouts at seed 6901. Parameter matching is necessary for pairing these records. Under the additional same-numerical-policy premise, Proposition 9 gives their complete-service adoption region, with normal-policy multiplier 504 and observed work differences. Cross-platform prediction portability is separately discussed in the supplement. The supplementary table reports the work savings and critical price ratios, including cases in which savings are negative. The loss margin is a prospective numerical design choice, not an estimated installation charge. Equivalence at that margin makes policy quality an eligible comparison; adoption depends on declared prices and complete work, including newly purchased qualification.

F.5. *Earlier evidence and independent reconstruction*

The original R67 catalogue remains a separate, unchanged companion, together with all preceding adverse studies. It records 212 returns, no neural complete-reuse winner, unresolved ordinary cost intervals, forty-eight neural fitting warnings and a severe horizon-four vector expansion. Those observations are not overwritten by the

new loss, new inference rows or different reuse window. The previous accumulation-aliasing counterexample and corrected execution also remain unchanged.

The new publication audit verifies every R69 receipt and stored-file identity, recomputes the achieved rational training certificates, checks every validation loss and executes independently written rational path calculations. A second interval implementation uses exact full-bin state and cost bounds, not only representative points, for unambiguous acquired paths. Its coverage and every diagnostic disagreement are recorded explicitly. This provides independent reconstruction of the sampled cash-path calculation and independent final inference arithmetic. It is not represented as an independent generation of every optimal-advantage endpoint or of the largest vector Bellman descendant tree.

APPENDIX G: ACHIEVED READOUTS AND INDEPENDENT VALIDATION

The two certificate-aware feature families use the same purchased states, action labels, endpoint enclosures, penalty and iteration budget. The replay verifies equality of every training-array payload at a fixed task and seed before comparing the methods. The label-loss readout shares the hat features but not the objective. The quadratic arm shares information but has a different parameter count. A finite-dimensional hat function has both a conventional piecewise-linear description and an exact ReLU realization; changing the encoding does not create a new independent method.

TABLE G.I: Achieved convex readout certificates

(d,T,r)	Rule	Seed	Weights	$F(w)$	$D(w)$	$F(w)/\tau^2$
2,2,2	B	1	10	0.0187	1.72e-08	7.35e+04
2,2,2	B	2	10	0.0161	2.56e-08	6.33e+04
2,2,2	H	1	10	0.0183	9.8e-08	7.19e+04
2,2,2	H	2	10	0.0154	3.26e-08	6.05e+04
8,3,2	B	1	40	0.00909	2.47e-12	1.89e+04
8,3,3	B	1	40	0.0112	0	2.34e+04
8,3,2	B	2	40	0.00891	5.94e-10	1.85e+04
8,3,3	B	2	40	0.00883	1.09e-09	1.84e+04

(d,T,r)	Rule	Seed	Weights	$F(w)$	$D(w)$	$F(w)/\tau^2$
8,3,2	H	1	40	0.00908	8.07e-10	1.89e+04
8,3,3	H	1	40	0.0112	0	2.34e+04
8,3,2	H	2	40	0.00882	8.81e-10	1.83e+04
8,3,3	H	2	40	0.00877	4.43e-10	1.82e+04

These are achieved objective and optimization-gap values, not optimizer flags or hidden-layer convergence claims. The complete rational fractions are retained in DETAILS70.json. Repeated services of the same readout are listed once here and remain separately charged in their receipts.

The objective and dual gap are evaluated on the returned quantized weights using rational contexts and features. The optimization certificate remains valid whether an approximate optimization routine reports success or a warning. The table concerns the achieved convex box problem, not unrestricted hidden-layer optimization. Raw weights, endpoints, labels and exact fractions are retained. Previously reported neural-fitting warnings belong to the unchanged earlier catalogue and are not removed by the successful computation of a new objective.

Exact context law and counting. Training and validation coordinates are uniform on the sixteen-bit dyadic grid, not on the continuous state cube or the full twenty-four-bit acquisition grid. Matching methods share these validation draws. The 6,144 method–context–date evaluations are not pooled as 6,144 independent states: every stated validation bound uses its own 256-row record. The separate cost-inference experiment retains the original continuous initial and innovation laws.

TABLE G.II: Independent-context work-loss validation

(d,T,r)	Rule	Seed	Returns	Root mean	Upper	All recovery queries
2,2,2	H	1	48	13.81	16.442	17680
2,2,2	H	2	50	13.68	16.315	17510
2,2,2	B	1	25	15.34	17.000	19635
2,2,2	B	2	21	15.61	17.000	19975
2,2,2	L	1	42	14.21	16.973	18190
2,2,2	L	2	47	13.88	16.482	17765

(d,T,r)	Rule	Seed	Returns	Root mean	Upper	All recovery queries
2,2,2	Q	1	45	14.01	16.707	17935
2,2,2	Q	2	52	13.55	16.282	17340
8,3,2	H	1	21	15.61	17.000	19975
8,3,3	H	1	23	15.47	17.000	462088
8,3,2	H	2	13	16.14	17.000	20655
8,3,3	H	2	21	15.61	17.000	469565
8,3,2	B	1	25	15.34	17.000	19635
8,3,3	B	1	23	15.47	17.000	462088
8,3,2	B	2	11	16.27	17.000	20825
8,3,3	B	2	20	15.67	17.000	471130
8,3,2	L	1	9	16.40	17.000	20995
8,3,3	L	1	6	16.60	17.000	486452
8,3,2	L	2	5	16.67	17.000	21335
8,3,3	L	2	13	16.14	17.000	481419
8,3,2	Q	1	8	16.47	17.000	21080
8,3,3	Q	1	21	15.61	17.000	464894
8,3,2	Q	2	10	16.34	17.000	20910
8,3,3	Q	2	4	16.73	17.000	493938

Each row has 256 new contexts; seed suffix 1 or 2 denotes 6901 or 6902. The upper bound is independently recomputed from an exact sample sum and an outward radius. It applies to the declared complete-action mesh recovery and context law, not automatically to an adaptive on-policy stream. Recovery-query totals include descendants.

The validation family contains twenty-four task–date–method–seed records with 256 contexts per record. The logarithm allowance twelve covers the stated finite family by a union bound. Reusing a context stream across methods induces dependence between estimators but does not invalidate that bound. The same stream is not counted as additional independent information for each method. The deterministic work inequality is checked on every context. The distributional upper endpoint is recomputed from the exact sample sum, an eighty-digit square root with an explicit expansion, and outward conversion to the printed numerical endpoint.

The full-action recovery mesh in this validation design has a declared root cap K. Its actual descendant queries are separately recorded. These observations do not validate a different adaptive fallback by definition. In Theorem 8, a distributional transfer allowance and a complete endpoint-work cap are additional premises. The theorem therefore handles predictable choices and dependent contexts without asserting that the recorded recursive state stream is independent.

APPENDIX H: PRECISION AND THE CONTINUOUS-LAW ESTIMAND

TABLE H.I

PAID INFERENCE WORK AND RAW ENCLOSURE COMPONENTS

(d,T)	Rule	Evaluator queries	Arm (s)	Conditional	State-bin	Ambiguous
2,2	A	101226	1.98	0.000113	2.3e-13	0
2,2	B	101226	1.91	0.000113	2.3e-13	0
2,2	H	101226	1.91	0.000113	2.3e-13	0
2,2	R	101226	1.83	0.000113	2.3e-13	0
8,3	A	2921243	100.74	0.000595	3.7e-13	0
8,3	B	2921234	102.56	0.000596	3.7e-13	0
8,3	H	2921234	102.59	0.000596	3.7e-13	0
8,3	R	2921465	101.36	0.000597	3.7e-13	0

Component columns are mean full widths before valid regret-range intersections. They do not add to the final narrower contrast width. Arm clocks are internal diagnostics; complete shared inference is charged as a whole: 9.03 seconds, 408.63 seconds. State-bin widths include propagated arithmetic. No acquisition ambiguity is discarded.

The centered observation is the sum of optimal Bellman advantages along an actual policy trajectory. Its expectation equals policy loss. Subtracting two such observations cancels the common initial optimal value in expectation. It does not cancel cash-flow risk path by path, and no empirical fitted value is substituted for the optimal value. Each conditional Bellman evaluation uses its own tolerance, action covering and continuous-innovation integration account, whose work is charged to the inference service.

The twelve paired intervals form one finite family over the two declared scalar tasks. With logarithm allowance twelve, $2 \cdot 12 \exp(-12) < 0.01$. For example, a rational partial sum of the exponential series already proves $\exp(12) > 2400$; no rounded logarithm is needed for the coverage claim. The ordinary cash-cost estimator uses a separate empirical-Bernstein family on the same rows. A joint claim about both families requires the sum of their tail budgets, not an assumption that the estimators are independent.

The mean conditional-value and state-bin component widths are recorded before intersecting with valid local regret ranges. The intersection can substantially reduce the resulting interval. Accordingly, those diagnostic widths need not sum to the final centered width, and they are not interpreted as independent errors. The state-bin column includes propagated state arithmetic. The actual final interval width is the mean final numerical width plus the two sampling radii, before any final support clipping.

For every path, the numerical action used by the controller is fixed before the more accurate evaluation. Its feasibility is checked on the true-state box as well as on the acquired state. An unresolved acquisition branch must receive a valid remaining support; it cannot be discarded. None is encountered in the new inference catalogue. The absence of sampled ambiguity is an observation, not a theorem excluding acquisition boundaries from the model.

APPENDIX I: INDEPENDENT FULL-BIN RECONSTRUCTION

The new independent implementation imports no NBO numerical kernel. It uses exact rational intervals for the continuous initial and innovation bins, twenty-four-bit acquired-state comparisons, true-state feasibility, stage costs, transitions and terminal costs. It treats the two cumulative endpoints as separate immutable rational numbers. In a transition coordinate, the deterministic function

$$f(u,v) = \frac{1}{16} + \frac{u}{2} + \frac{v}{8} + \frac{u(1-v)}{16}$$

has derivatives $1/2 + (1-v)/16 \geq 0$ and $1/8 - u/16 \geq 0$ on the state cube. Thus evaluation at the lower–lower and upper–upper corners bounds its range without

invoking the production interval library. Quadratic costs use exact interval squares; shortage uses the monotone positive-part map followed by an interval square. Both shocks keep their original widths and marginal laws.

The implementation checks that its complete bin enclosure lies inside the archived cash-cost enclosure. A second, independently written rational midpoint calculation checks acquisition and cumulative cost at a represented point. The midpoint test is not substituted for the full-bin test. Across 3,706 physical path archives there are 38,722 recorded path rows; 736 distinct payloads contain 14,242 unique rows. Repeated timing copies are hash-checked and compared, but their duplicate rational evaluation is not counted again. The independent full-bin replay covers all 14,242 unique rows, with no unresolved branch in this executed catalogue.

A separate exact-sum and high-precision calculation checks every final centered confidence interval from its stored advantage endpoints. This independently verifies final inference arithmetic, not the generation of those optimal-advantage endpoints. The regression suite compares primitive identities and small complete value problems with separate rational formulas. It does not independently regenerate the largest multivariate Bellman descendant tree. The release states these different coverage levels rather than assigning an undifferentiated independent-verification label to the whole program.

I.1. *Prediction portability and retained disagreements*

An initial strict replay stopped because one Bernstein validation prediction was not bitwise equal to a recomputed value. That failed attempt, its original audit source and traceback are retained on their own branch and in the publication inputs. The failure does not by itself identify an invalid Bellman bound. Nor does a successful rerun on another processor prove the earlier numerical result bitwise reproducible.

The current audit checks the saved actual action on its declared lattice and recomputes its strong-convexity chord with exact rational arithmetic. A discrepancy with a recomputed readout is additionally checked against the declared action-rounding and prediction allowances, not accepted by an arbitrary numerical closeness thresh-

old. The original scientific endpoints, actions, models and source freeze are unchanged. Numerical parameter identity, callable-policy identity and successful certificate verification are distinct statements.

In particular, equality of saved readout arrays is necessary but not sufficient for transferring a cost estimate to an implementation on a different arithmetic runtime. The complete-service purchase region assumes the same numerical callable policy or an additional valid implementation-transfer bound. Matching processor metadata or passing a finite replay is not silently promoted to an all-state policy-identity theorem.

APPENDIX J: COMPLETE WORK AND CONDITIONAL ADOPTION

TABLE J.I: Conditional reuse-stage adoption thresholds

(d,T)	Rule	Repeat	Saving (s)	Loss upper	p/v threshold
2,2	H	0	0.000	0.3441	1383.7120
2,2	H	1	0.350	0.3441	0.9821
2,2	H	2	-0.201	0.3441	–
2,2	H	3	3.457	0.3441	0.0996
2,2	B	0	-0.201	0.3440	–
2,2	B	1	0.601	0.3440	0.5727
2,2	B	2	-0.401	0.3440	–
2,2	B	3	3.808	0.3440	0.0904
8,3	H	0	-1.153	0.8079	–
8,3	H	1	-1.301	0.8079	–
8,3	H	2	-1.399	0.8079	–
8,3	H	3	-1.555	0.8079	–
8,3	B	0	-1.306	0.8079	–
8,3	B	1	-1.606	0.8079	–
8,3	B	2	-1.302	0.8079	–
8,3	B	3	-1.955	0.8079	–

Seed 6901; zero incremental installation fee and common or previously paid qualification. The loss upper is 504 times the corresponding policy-cost upper endpoint. A displayed positive-saving threshold describes the adoption region only when the numerical callable policy is the one evaluated by inference. Saved parameter equality is checked, but cross-platform numerical equality is an additional requirement. A dash means no positive work saving in that matched observation. Newly purchased qualification work must be added before claiming a first-purchase saving. Prices are declared inputs, not estimated currency values.

Qualification is not free. The displayed reuse thresholds condition on a common or previously paid qualification calculation. The process receipts cover the complete reuse service, not the separate policy-cost inference experiment. For a first purchase, let C_q be additional qualification work not paid by the benchmark alternative. In Proposition 9, use $W_\pi = W_\pi^{\mathrm{run}} + C_q$ and $W_\nu = W_\nu^{\mathrm{run}}$; the observed saving becomes $S - C_q$. This is a change of the explicit accounting boundary, not an additional theorem assumption about free inference. Charging the entire observed shared inference process gives C_q approximately 9.03 seconds in the small task and 408.63 seconds in the larger task. Every displayed matched saving then becomes negative. Thus the current receipts do not identify a positive qualification-plus-reuse work saving under that conservative first-purchase boundary. An allocated marginal inference time or a longer repayment horizon was not separately executed. The machine-readable table retains both boundaries, and the general price region remains valid with the actual charged work and installation fee.

The multiplier 504 follows from sixty-four blocks of eight trajectories with one predetermined fallback block. In that block the service uses the benchmark itself, giving zero expected policy-cost difference. In the other blocks it uses the fixed candidate. The cost interval can therefore be scaled by 504 under the policy-identity premise of Proposition 9. A positive work saving is an observed matched process difference, not a confidence bound for future latency. The complete timing range also records all unfavorable repetitions and does not authorize selection of a favorable timing result as a population estimate.

The parent receipt includes child launch, imports, fitting or loading, all policy and path calculations, serialization, child exit and log flushing. Parent receipt

bookkeeping belongs to the group clock. Training, inference and publication replay have different boundaries and are not assigned zero cost when excluded from a displayed sub-account. The machine record includes the permitted CPU, thread counts, available frequency and governor data. Processor frequency was not locked by the experiment. Fine timing differences consequently remain descriptive even when operation counts reproduce exactly.

The original normalized investment units are retained. The loss margin is fixed by the numerical protocol; the value of a loss unit, compute price and installation fee are purchaser inputs. This produces a transparent conditional adoption region, not an estimated monetary welfare result. Its noninferiority and equivalence statements concern the actual expected-cost target and do not imply strict neural superiority.

APPENDIX K: PRESERVATION AND SCOPE

Every active R67 theorem statement is retained verbatim in the current theoretical sections, and every proof is retained in the article or this supplement. The original empirical article and its full supplement are unchanged companion documents. They retain all earlier conventional winners, unresolved direct-cost comparisons, fitting warnings, vector horizon costs and the accumulation-aliasing counterexample. The broader NBO development remains available under its original diffusion, recursive-preference, temporal-self and game hypotheses; those hypotheses are not replaced by the convex expected-cost premises of the current query theorem.

The new frozen R69 catalogue is integrated once. Publication replay, repeated clocks, repeated random streams, duplicate path payloads and regenerated tables do not increase its number of independent economic observations. The final source and content map records the location and hash of each retained item. Compilation, regression tests and independent numerical calculations are evidence for their stated checks, not an external editorial decision or a formal proof-checker certificate.

TABLE L.I

COMPLETE COLD-START SERVICES AT A COMMON INFORMATION BUDGET

(d, T)	Rule	Processes	Policy queries	Median (s)	Range (s)
2,2	A	4	597	0.365	[0.364, 0.366]
2,2	B	8	607–612	0.465	[0.465, 0.466]
2,2	H	8	592–612	0.465	[0.465, 0.466]
2,2	L	8	592–597	0.440	[0.414, 0.466]
2,2	Q	8	597	0.415	[0.415, 0.967]
2,2	U	4	597	0.390	[0.364, 0.415]
8,3	A	4	2594	0.666	[0.615, 0.666]
8,3	B	8	2594	1.618	[1.618, 1.619]
8,3	H	8	2594	1.618	[1.618, 1.669]
8,3	L	8	2594	1.518	[1.517, 1.568]
8,3	Q	8	2594	1.518	[1.467, 1.519]
8,3	U	4	2594	0.641	[0.615, 0.666]

A: adaptive; U: upper-chord minimizer; H: certificate-aware hat–ReLU; B: certificate-aware Bernstein; L: label-loss hat–ReLU; Q: quadratic. Queries concern the policy workload; training and validation queries are additional. Four process repetitions per seed are descriptive clocks, not independent fits or confidence bounds. All complete times include launch and shutdown.

APPENDIX L: COMPLETE COLD-START AND VECTOR WORKLOADS

The tables retain every declared method and task. They are relocated intact from the article to keep its reading path focused; no row, adverse comparison or timing repetition is removed.

APPENDIX M: NEW REUSE RECORDS AND INDEPENDENT ARITHMETIC

The new scientific source commitment binds all five new scientific modules and their inherited Python dependencies, together with the exact task and timeout catalogue. It precedes execution. The numerical study contains no newly fitted predictor and no new independent policy-cost sample. Its roots are predetermined localized states, not a uniform or stationary deployment distribution.

TABLE L.II

TWO-CONTROL AND TWO-INNOVATION COMPLETE SERVICES

(d, T)	Rule	Processes	Queries	Median (s)	Peak KiB
2,2	H	2	98	0.47	65520
2,2	Q	2	99	0.49	65524
2,2	S	2	82	0.39	65164
2,2	U	2	82	0.36	65328
2,3	H	2	5634	5.83	66004
2,3	Q	2	5914	5.93	66256
2,3	S	2	4541	3.70	65184
2,3	U	2	9039	7.93	65016
2,4	H	1	872033	62.01	65872
2,4	S	1	592658	41.37	65256
8,2	H	2	91	0.46	65780
8,2	Q	2	91	0.46	65864
8,2	S	2	75	0.36	65252
8,2	U	2	75	0.36	65336
8,3	H	2	3495	5.93	66028
8,3	Q	2	3495	5.85	66208
8,3	S	2	2645	3.55	65168
8,3	U	2	3278	4.60	65544

S: simplicial; U: largest-edge uniform refinement; H and Q: trained two-output hat and quadratic proposals. Ordinary services have eight continuous-bin paths. Horizon-four services have one path, and H transfers its three-date readout. These are workload records, not precision-designed estimates of population policy costs. Peak memory is the maximum observed resident set within the listed processes.

TABLE M.I: Every new complete-process repetition

Task	Rule	Rep.	Seconds	Queries	Hits	RSS
S2/2	Fresh	1	1.166251	490	0	97,804
S2/2	Exact	1	0.715207	115	25	97,944
S2/2	Geometric	1	0.665186	55	29	97,808
S2/2	Exact	2	0.765338	115	25	97,816

Task	Rule	Rep.	Seconds	Queries	Hits	RSS
S2/2	Geometric	2	0.668814	55	29	97,816
S2/2	Fresh	2	1.216814	490	0	97,804
S2/2	Geometric	3	0.668796	55	29	97,800
S2/2	Fresh	3	1.115930	490	0	97,928
S2/2	Exact	3	0.765269	115	25	97,932
S8/3	Fresh	1	2.739717	1,610	0	97,936
S8/3	Exact	1	1.921036	959	7	97,944
S8/3	Geometric	1	1.115856	401	13	97,936
S8/3	Exact	2	1.867363	959	7	97,932
S8/3	Geometric	2	1.216160	401	13	97,936
S8/3	Fresh	2	2.772324	1,610	0	97,932
S8/3	Geometric	3	1.166799	401	13	97,932
S8/3	Fresh	3	2.721673	1,610	0	97,944
S8/3	Exact	3	1.867344	959	7	97,932
V2/2	Fresh	1	2.422182	397	0	97,932
V2/2	Exact	1	1.219324	125	17	97,932
V2/2	Geometric	1	0.915594	61	21	97,932
V2/2	Exact	2	1.216086	125	17	97,944
V2/2	Geometric	2	0.919073	61	21	97,936
V2/2	Fresh	2	2.470714	397	0	97,936
V2/2	Geometric	3	0.915826	61	21	97,928
V2/2	Fresh	3	2.471993	397	0	97,932
V2/2	Exact	3	1.216277	125	17	97,936
V2/3	Fresh	1	21.474661	25,534	0	98,072
V2/3	Exact	1	14.203954	17,023	3	98,076
V2/3	Geometric	1	9.388726	11,349	5	98,056
V2/3	Exact	2	14.556513	17,023	3	98,056
V2/3	Geometric	2	9.793062	11,349	5	98,008
V2/3	Fresh	2	22.620244	25,534	0	98,064
V2/4	Fresh	1	347.971704	5,085,947	0	99,212
V2/4	Exact	1	168.526652	2,542,973	2	99,216
V2/4	Geometric	1	84.068560	1,271,487	3	99,212

Each row is one separately launched and completed process. No clock repetition is a new fitting seed or an independent policy-cost observation. All stage and source identities remain in its receipt.

TABLE M.II: Paid fills, rejected records and lookup overhead

Task	Rule	Rep.	Checks	Stored	Stale	Fill (s)	Lookup (s)
S2/2	Fresh	1	0	0	0	0.5554	0.0000
S2/2	Exact	1	25	4	3	0.1301	0.0055
S2/2	Geometric	1	30	2	1	0.0642	0.0039
S2/2	Exact	2	25	4	3	0.1349	0.0056
S2/2	Geometric	2	30	2	1	0.0641	0.0040
S2/2	Fresh	2	0	0	0	0.5678	0.0001
S2/2	Geometric	3	30	2	1	0.0621	0.0041
S2/2	Fresh	3	0	0	0	0.5363	0.0000
S2/2	Exact	3	25	4	3	0.1344	0.0051
S8/3	Fresh	1	0	0	0	2.1140	0.0000
S8/3	Exact	1	7	5	4	1.2850	0.0036
S8/3	Geometric	1	14	2	1	0.5343	0.0029
S8/3	Exact	2	7	5	4	1.2600	0.0039
S8/3	Geometric	2	14	2	1	0.5502	0.0029
S8/3	Fresh	2	0	0	0	2.1810	0.0000
S8/3	Geometric	3	14	2	1	0.5306	0.0033
S8/3	Fresh	3	0	0	0	2.1113	0.0000
S8/3	Exact	3	7	5	4	1.2652	0.0038
V2/2	Fresh	1	0	0	0	1.8126	0.0001
V2/2	Exact	1	17	4	3	0.5964	0.0040
V2/2	Geometric	1	22	2	1	0.3040	0.0032
V2/2	Exact	2	17	4	3	0.6196	0.0042
V2/2	Geometric	2	22	2	1	0.3143	0.0033
V2/2	Fresh	2	0	0	0	1.8519	0.0000
V2/2	Geometric	3	22	2	1	0.3170	0.0033
V2/2	Fresh	3	0	0	0	1.8495	0.0000
V2/2	Exact	3	17	4	3	0.6215	0.0041
V2/3	Fresh	1	0	0	0	20.8660	0.0000

Task	Rule	Rep.	Checks	Stored	Stale	Fill (s)	Lookup (s)
V2/3	Exact	1	3	3	2	13.5845	0.0009
V2/3	Geometric	1	6	2	1	8.7833	0.0013
V2/3	Exact	2	3	3	2	13.9392	0.0011
V2/3	Geometric	2	6	2	1	9.1646	0.0012
V2/3	Fresh	2	0	0	0	21.9832	0.0000
V2/4	Fresh	1	0	0	0	347.3480	0.0000
V2/4	Exact	1	2	2	0	167.8846	0.0006
V2/4	Geometric	1	3	1	0	83.4527	0.0006

Stored is the maximum number of retained witnesses. Stale counts discarded records at the declared context invalidation. Fill and lookup are internal subclocks, not the complete process boundary. The lookup account includes record-load validation. Repeated storage, hashing, output and shutdown are additionally included in complete receipts.

M.1. *Complete timing and retained records*

The process receipt includes creation, Python imports, original Bellman fills, all recursive descendants, cache retrieval, rational checks, repeated serialization, final output and process exit. The driver writes a separate receipt after that boundary. Summing child block clocks is not the complete process time. Every block's body prefix excludes imports and has its own exact definition in the source. The complete receipts, rather than those prefixes, determine the reported timing comparison.

No processor frequency lock is claimed. The start and end records include the selected CPU, frequency fields where the environment exposes them, interpreter version and platform. A cyclic mode order limits a simple fixed-order confound but does not create randomized hardware-frequency inference. Close clock differences require more evidence than the prescribed finite repetitions. Bellman action counts are consequently reported separately from seconds. They are invariant descriptive counts for these transcripts, not predictions of another implementation's throughput.

The 128-record cap bounds logical retained witnesses. It does not bound total process resident memory: a fresh recursive fill can retain batched descendants,

interpreter state, interval work arrays and output buffers. The largest process memory remains in the table. The executed cache operates at workload roots only. The four-date pressure is not hidden by counting a later root hit as though the first fill used no descendants.

M.2. *Independent checks and trusted premises*

The additional admission audit matches all 708 returns to the prescribed state, control dimension, horizon, tolerance and context epoch. Cache reloads in this catalogue use the same process and numerical model. Arbitrarily supplied external intervals are not made valid by deserialization or a file hash.

For a returned hit, the independent checker locates the earlier paid source record with the same context, horizon, source state, action and endpoints. It recomputes the original regularity constants, state distance, action displacement and interval expansion in exact rational arithmetic. It checks the returned action's lattice and capacity and that the serialized binary endpoints contain the exact expanded interval. A narrow stored interval alone would not prove that a source fill encloses the original value: the inherited verifier and its primitive proof remain explicit premises.

For conditional ReLU integration, the second implementation reconstructs the original projected drift and shocks independently. It does not call the divided-power formula. It clips the support rectangle against the affine neuron's positive half-plane and uses oriented area and first moments. Degenerate projections reduce to one-dimensional integration or a deterministic positive part. All rational discrepancies are fatal to the new arithmetic audit and are retained as evidence if encountered.

The earlier R70 independent cash-path audit and final inference arithmetic are unchanged. The historical full descendant trees are not certified by pretending that a file hash is a mathematical proof. Neither the new rational transfer checks nor the polygon integrations independently regenerate all those trees. The release states this boundary in its machine-readable audit and response.

M.3. *Development probe and publication preservation*

Before the new source commitment, a vector development probe queried $(0.4, 0.6)$ at tolerance 0.04 and horizons two, three and four. The first two returned. The probe was stopped at its 35-second process limit while computing horizon four. Its log is retained separately and is not merged with the later 900-second scientific catalogue or used as a favorable timing observation.

Every R70 active section is assigned to exactly one current document and checked byte for byte. All inherited formal statements and proofs are thereby retained, not reconstructed from a favorable summary. The complete old main article and supplement remain companion PDFs. New tables are derived from the committed receipts and independent audit, and all four current documents are rebuilt from a clean source directory. Visual inspection is documented against the exact PDF hashes actually inspected. Compilation and tests do not constitute external mathematical acceptance.

APPENDIX N: COMPLETE CONTINUATION-QUALIFICATION RECORDS

The fixed design was committed before any new construction service ran. Its four points are the product of covers with 8^2 or 16^2 state nodes and two nine-coefficient bases. The terminal continuation is the original g; each preceding date is fitted against the continuation actually returned at the following date. The stage label is an interval containing the full continuous-action Bellman infimum. The coefficient box, quantum, feature knots, action-cover size and optimization tolerance were fixed in advance.

The ReLU basis consists of the constant and, in each coordinate, the identity and positive parts at $1/4$, $1/2$ and $3/4$. The conventional basis consists of the constant and the four nonconstant degree-four Bernstein polynomials per coordinate. Their valid feature moduli are respectively one and four. Having the same number of coefficients does not make their function classes or approximation moduli identical. Each basis receives its own actual fixed-future labels and the same construction rule; the label-loss ablation within each stage shares those exact labels and features.

TABLE N.I

EVERY ATTAINED READOUT AND ITS FIXED-LABEL LEAST-SQUARES ABLATION

n	Basis	Date	$\mathcal{W}$ upper	LS bound upper	Dual gap
8	ReLU	0	2.38478	3.59860	0.000000007
8	ReLU	1	3.21372	4.91701	0.000000014
8	Bernstein	0	1.92163	5.25637	0.000000000
8	Bernstein	1	3.67363	9.63856	0.000000000
16	ReLU	0	1.63091	1.99849	0.000000014
16	ReLU	1	2.24222	2.75521	0.000000014
16	Bernstein	0	1.59492	2.78996	0.000000000
16	Bernstein	1	3.02915	5.10081	0.000000000

LS is bounded least squares on the identical features and exact fixed-future interval midpoints. Its bound is evaluated by the same residual-qualification formula. It is not a separately propagated controller. The primal and dual calculations are exact rational checks, independently reconstructed without relying on the optimizer status.

Every linear-program output is rounded into its declared coefficient box and checked with rational probability multipliers. The table's gap is the achieved primal value minus the exact dual lower bound, not the solver's reported duality tolerance. Constant recentering changes the continuation gauge without changing this objective. The predeclared success tolerance is $1/65536$. The stored status and message remain even when the exact certificate succeeds; an unsuccessful optimizer would retain its feasible fallback and its actual gap rather than disappear from the catalogue.

The service clock includes both backward constructions, all exact action and feature-mean evaluations, the qualification optimizer, its rationalization, the matched bounded least-squares ablation and the service output. The separate clock record identifies its endpoint. These are single recorded construction times; no estimate of mean runtime, clock stability or a universal neural speed ratio is inferred. The independent audit has its own time and never replaces a scientific service time. Exact arithmetic-operation counts are not a bound on rational bit complexity.

TABLE N.II

COMPLETE NEW CONSTRUCTION WORK AND OBSERVATION SCOPE

n	Basis	Action values	Feature means	Seconds	Cost rows
8	ReLU	8448	38016	1.174	0
8	Bernstein	8448	38016	3.174	0
16	ReLU	33792	152064	4.793	0
16	Bernstein	33792	152064	10.617	0

Action values include continuous-action covering and whole-cell feasible proposals at both dates. Feature means count evaluations for the fitted future, not terminal formulas. Seconds include both fits, qualification, matched least-squares ablations and the service output; the independent checker has a separate clock. There are no recursive optimal-value queries and no new independent policy-cost observations. Single-service times are descriptive.

The independent action-value evaluator does not invoke the production positive-part moment expansion. It partitions the innovation interval at every ReLU breakpoint and integrates affine pieces. For the degree-four conventional continuation it uses the positive Boole weights on five equally spaced points; the result is exact for this polynomial degree. Final-date values are checked against the retained original rational investment primitive. It also reconstructs every feasible node action, the original continuous-action lower enclosure, the robust acquired action and its whole-cell gap. Thus the newly computed residual account is tied to original primitives rather than only to a stored hash.

The current construction does not sample a new expected policy-cost difference. The support supplied by its fitted continuation is available to Theorem 11, but no new stopped investment comparison is asserted. The separately executed fixed-sample comparisons, two-control workloads and cache processes retain their original observations, contracts and coverage accounts.

APPENDIX O: EXACT SEQUENTIAL-LEDGER REGRESSIONS

The new regression suite checks conditional means, range reservation, target changes and failed numerical episodes. Its adaptive binary-tree diagnostic enumerates all $2^{12} = 4096$ leaves and every prefix, with exact probabilities and changing

predictable ranges. The event count is a finite-model check of the implemented boundary, not an empirical substitute for the supermartingale proof. Rational logarithm bounds and coefficient duals are also tested independently of floating solver stopping flags.

A support failure and a numerical failure are deliberately different branches. An empty intersection or a changed fixed target is rejected. A numerical failure records the prequalified full support, advances the prefix and charges its work. The interface cannot prove that a caller fixed the support before observing the episode; that temporal and probabilistic premise remains part of the algorithm's contract. A cache carried across episodes is included in the target definition or is handled as varying conditional cost.

All prior scientific arrays and sources remain unchanged. The ordinary-source build runs the inherited and current tests and recomputes the published tables. The clean archive includes the transitive R62 input sources, including the original service kernel omitted by the interrupted R72 archive. The reproduction report identifies the suites run inside that archive and compares actual PDF text and page counts after deleting generated PDFs and tables. Visual inspection is recorded separately against the output hashes. None of those publication checks is an external mathematical acceptance decision.

APPENDIX P: TRANSPORTING PURCHASED BELLMAN WITNESSES

A second way to control the descendant cost is to reuse a calculation already purchased. Exact memoization only recognizes the same state. The following rule transfers a lower Bellman bound and its feasible upper witness to a nearby state, after repairing capacity and accounting for the full displacement. It is compatible with either a neural or a conventional generator of the original certified record.

P.1. *An original-optimum transport certificate*

At remaining horizon r, a record at y contains an action $a_y \in A(y)$ and numbers satisfying

$$L_y \le V_r^*(y) \le Q_r^*(y, a_y) \le U_y.$$

It also identifies the economic primitives, remaining horizon and numerical contract under which the record was obtained. A file hash detects changed bytes; validity of the inequalities is a separate premise. For the original scalar and two-control economies, let

$$L_V = G_r, \qquad L_Q = \frac{8}{d} + \frac{11\beta}{16} G_{r-1}, \qquad L_A = \frac{13}{16} + \frac{3\beta d}{8} G_{r-1}.$$

These bound state variation of V_r^* and state and action variation of Q_r^*. The action comparison may be made on the ambient simplex $a \geq 0$, $\sum_i a_i \leq 1/4$: the original transition remains in the state cube there, even if a particular action violates the smaller capacity at one endpoint.

At a new state x, repair a_y to an actual lattice action $\widetilde{a} \in A(x)$. Set $D = \|x - y\|_1$ and $\Delta_a = \|\widetilde{a} - a_y\|_1$. Define

$$L(x) = \max\{0, L_y - L_V D\}, \qquad U(x) = U_y + L_Q D + L_A \Delta_a. \tag{13}$$

The truncation at zero uses the nonnegative costs of this economy. It is not part of a general signed-cost transfer rule.

THEOREM 10—Reuse after capacity repair: *Under the stated record and modulus conditions,*

$$L(x) \leq V_r^*(x) \leq Q_r^*(x, \widetilde{a}) \leq U(x).$$

Hence outward endpoints with $\overline{U}(x) - \underline{L}(x) \leq \delta$ permit the actual repaired action to be returned without a new Bellman query. An identity mismatch, failed feasibility check or excessive width requires a new verified calculation; proximity alone licenses no reuse.

The same assertion holds after any history-dependent selection of valid records. At each future date, suppose the selected record or fresh query supplies the original local-gap and acquisition premises of Theorem 14. The resulting, possibly history-dependent controller retains that theorem's original-optimum loss sum. It need not coincide with the action returned by a fresh search at the same tolerance.

The proof is in Supplement, Section 5.3.

The implemented repair first rounds the new capacity downward on the 2^{-32} lattice, scales a vector action proportionally when necessary, and rounds each

coordinate downward. The certificate charges its exact resulting ℓ^1 displacement, not merely the change in capacity. Retrieval may use approximate distances or a learned index as an ordering heuristic. Every returned hit must still satisfy (13) under outward arithmetic.

P.2. *A finite-cover account and its premises*

A small number of hits does not establish a uniform reuse complexity bound. Such a bound requires a fill precision, a coverage argument and a retention rule. Let the action quantum be q, let $m \in \{1,2\}$, and suppose every fresh record is purchased with gap at most $\gamma\delta$, where $0 < \gamma < 1$. The repair just described obeys

$$\Delta_a \le \kappa D + (m+1)q, \qquad \kappa = 1/(8d).$$

Consider the exact-endpoint transport rule, or an outward rule with a known total additional width allowance $\xi \ge 0$. Define

$$\rho_* = \frac{(1-\gamma)\delta - L_A(m+1)q - \xi}{L_V + L_Q + \kappa L_A}. \tag{14}$$

PROPOSITION 11—Paid fills and complete reuse work: *Suppose $\rho_* > 0$, all valid records are retained, the context is fixed, and every stored candidate is checked before a miss is declared. A record within distance ρ_* certifies a hit. On the unit state cube, the number M_r of fresh records at remaining horizon r is at most*

$$M_r \le \left(1 + \frac{d}{\rho_*}\right)^d. \tag{15}$$

If $C_r(\gamma\delta)$ bounds the complete cost of a fresh fill, a workload of B requests obeys

$$W_B \le W_{\text{setup}} + M_r C_r(\gamma\delta) + \sum_{b=1}^{B} \left(W_b^{\text{lookup}} + W_b^{\text{check}} + W_b^{\text{load/store}}\right). \tag{16}$$

The lookup and check costs include rejected candidates. Naive retrieval uses $O(M_r d)$ arithmetic operations per request, apart from rational bit costs. Storage, eviction, invalidation and context changes require their own additional fill accounts when the stated retention premises do not hold.

The proof is in Supplement, Section 5.4.

This bound makes the dimensional price of uniform reuse explicit. It is not a dimension-free caching theorem. It also separates a first expensive Bellman fill from later transport checks. Caching at the workload root can avoid all descendants on a hit, but does not accelerate the descendants of a miss. The executed bounded cache uses 128 records and the requested fill tolerance, not the tighter $\gamma\delta$ premise. Its operation counts below are direct observations, not an application of (15). A cache at recursive descendants would need a separate execution and ledger.

APPENDIX Q: CONTINUATION EVALUATION AND PAID QUERY REUSE

The evidence separates three questions: whether a certified readout returns an admissible policy, whether its actual expected cost meets a declared margin, and whether a purchased calculation can be reused profitably. The preceding training and inference catalogue remains unchanged. The new execution fixes localized query workloads and exact conditional-moment checks before running them. It changes neither the economic primitives nor the continuous feasible-action target.

Q.1. *Retained trained policies and economic precision*

The retained 188-process catalogue is reproduced in Appendix, Section F. Hat–ReLU and Bernstein readouts purchase the same teacher information and have exactly $5d$ coefficients; the achieved objective and box dual gaps are recomputed rationally. Independent validation and sixty-four-block reuse receive separate work accounts. Both their favorable and unfavorable repetitions remain visible. In particular, the eight-state task does not identify a learned complete-work winner.

The direct policy-cost experiment fixes its sample count and numerical precision before execution. All twelve original-law paired intervals lie within the declared equivalence region $[-1/256, 1/256]$, and all contain zero. The two adaptive-minus-hat–ReLU intervals are approximately $[-0.00068263, 0.00070764]$ and $[-0.00160291, 0.00160253]$. Their shared inference services cost about 9.03 and 408.63 seconds. These are informative quality comparisons with expensive qualifi-

cation, not an identified neural cost ranking or a free inference improvement. The complete precision and adoption tables are retained in the appendix.

The frozen-continuation theorem supplies a different way to evaluate that expected-cost estimand. Its moment component is executed here; a new trained critic meeting the old equivalence margin at lower complete qualification cost is not presumed. The earlier optimal-advantage observations are not counted again as samples of the new formula.

Q.2. *A fixed original-law query catalogue*

The new design has five tasks: two scalar-control tasks at $(d,T) = (2,2),(8,3)$ and three two-control tasks at $(d,T) = (2,2),(2,3),(2,4)$. Each task has three rules: fresh calculation, exact-key reuse and geometric witness transport. The source commitment fixes the task list, tolerances, repetition counts, 128-record capacity, invalidation schedule, workload order and 900-second per-process limit. There are 36 prescribed complete processes. None trains a new action predictor or loads a precomputed conventional policy.

A workload starts at alternating coordinates $0.4, 0.6$. Odd requests repeat that state. Predetermined even requests perturb one coordinate by 2^{-20}. Except in the four-date stress case, a late distant request has every coordinate equal to 0.75, and a specified earlier block invalidates the context. The first four tasks have 32, 16, 24 and 8 requests; the four-date task has four. The first three tasks have three timing repetitions, the three-date vector task two, and the four-date task one. Rules are executed in a fixed cyclic order across repetitions.

Every miss calls the unchanged original recursive verifier at the declared tolerance. Every hit uses the exact state-transfer and capacity-repair test. The cache is serialized, hashed, reloaded and validated after every block. The receipt begins before process creation and ends after exit, including imports, all descendant work, file output and shutdown; writing the parent receipt itself is separate publication overhead. A single CPU is pinned. Available start and end frequency information is recorded, but frequency is not locked. Repetitions support descriptive comparisons, not a hardware-uniform or population timing theorem.

TABLE Q.I

COMPLETE ORIGINAL-LAW QUERY WORKLOADS WITH AND WITHOUT REUSE

Task	Rule	Fills	Hits	Bellman queries	Seconds
S2/2	Fresh	32	0	490	1.166
S2/2	Exact	7	25	115	0.765
S2/2	Geometric	3	29	55	0.669
S8/3	Fresh	16	0	1,610	2.740
S8/3	Exact	9	7	959	1.867
S8/3	Geometric	3	13	401	1.167
V2/2	Fresh	24	0	397	2.471
V2/2	Exact	7	17	125	1.216
V2/2	Geometric	3	21	61	0.916
V2/3	Fresh	8	0	25,534	22.047
V2/3	Exact	5	3	17,023	14.380
V2/3	Geometric	3	5	11,349	9.591
V2/4	Fresh	4	0	5,085,947	347.972
V2/4	Exact	2	2	2,542,973	168.527
V2/4	Geometric	1	3	1,271,487	84.069

S and V denote scalar and two-control tasks; numbers are state dimension and horizon. Queries include every recursive descendant. Seconds are medians of complete processes: three repetitions for S2/2, S8/3 and V2/2, two for V2/3, and one stress observation for V2/4. All prescribed records are retained. These are localized workload comparisons, not population policy-cost samples.

The table counts Bellman action evaluations at all descendants, not only workload roots. Geometric hits at a nonidentical state distinguish transport from exact memoization. Their capacity-repair count is separately recorded, including downward repairs that change the returned action. The distant request and context invalidation test whether a cheaper rule refuses an unjustified reuse rather than forcing a hit. An unchanged numerical law with a new context identifier still triggers the declared conservative invalidation.

The four-date task retains the first expensive tree. A hit avoids that tree on a later request, whereas a miss pays for it again. This is a local repeated-query

TABLE Q.II

FOUR-DATE, TWO-CONTROL FIRST-FILL AND COMPLETE WORKLOAD ACCOUNT

Rule	First-fill queries	All queries	Seconds	RSS (KiB)
Fresh	1,271,487	5,085,947	347.972	99,212
Exact	1,271,487	2,542,973	168.527	99,216
Geometric	1,271,487	1,271,487	84.069	99,212

Four requested root states, two independent innovations, and common local tolerance 512/14911. The first-fill column counts its full descendant tree. Exact reuse recognizes identical states; geometric reuse additionally transfers a valid witness to a nearby state. RSS is total process peak memory, not logical cache bytes. One repetition is descriptive only.

improvement under the original optimum contract, not a disappearance of the long-horizon first-fill cost. The trace records every source witness, state displacement, repaired action and outward endpoint. The absence of a state lattice does not justify omitting either the descendant count or resident memory.

The full repetition and storage ledger is in Appendix, Section M. Prefix clocks there begin after module loading and are labeled as body clocks; they are not substituted for complete process receipts. The bounded cache and requested fill tolerance do not satisfy the stronger retained-cover premises of Proposition 11, so that proposition's uniform packing count is not presented as an empirical result.

Q.3. *Independent conditional-moment and transport checks*

The rational integration catalogue uses four (d, m) specifications, namely $(2,1), (8,1), (2,2), (8,2)$, and 64 state–action contexts per specification. Each context has sixteen rational neurons. The original bilinear state drift, action exposure and uniform innovation projections are evaluated exactly. An independent checker clips the corresponding innovation rectangle and integrates the affine neuron through polygon area and first moments. It imports no production moment, cache or interval-query routine.

The same independent checker verifies every new hit from its earlier paid fill, including feasibility, action lattice, context, original primitive moduli, exact trans-

TABLE Q.III

INDEPENDENT CHECKS OF THE NEW OPERATIONS

Quantity	Count
Prescribed complete processes	36
Returned / failed	36 / 0
Checked root returns	708
Checked reuse returns	357
Different-state reuse returns	127
Repaired-action reuse returns	18
Independent exact neuron integrals	4096
New independent policy-cost rows	0

The checker imports no production numerical kernel. It verifies each new transfer against a prior paid fill and independently integrates each rational neuron on the original innovation support. It does not independently regenerate the internal optimal-value trees of inherited fills.

port endpoints and outward conversion. This is independent reconstruction of the new transfer operation. The inherited recursive fill endpoints remain a disclosed dependency; their entire internal trees are not independently regenerated by this checker. The preceding independent full-bin cash-path calculations and the historical accumulation-aliasing counterexample remain in the retained appendix and companions.

The moment tests establish exact conditional integration, not the accuracy of an unobserved fitted critic. The query workload establishes a localized certificate-preserving reuse comparison, not independent expected policy-cost observations. A history-dependent transported policy is not identified with the frozen policies in the old cost-inference study. Its economic adoption therefore requires its own eligible cost account or a decision maker's declared worst-case accuracy requirement, together with the measured fill, reuse and qualification charges.

Q.4. *Constructed continuations and attained qualification*

The readout theorem is executed in four predeclared backward constructions in the original two-state, two-date, scalar-control economy. Two midpoint covers, with

eight or sixteen cells per coordinate, are crossed with additive shallow ReLU and degree-four Bernstein features. Each basis has nine coefficients bounded by eight. At each date the future is the continuation actually constructed at the next date. Exact conditional moments supply thirty-three continuous feasible action values per node and thirty-three robust, downward-quantized acquired-action proposals. The original state-dependent capacity and continuous innovation law are unchanged.

TABLE Q.IV

ATTAINED CONTINUATION QUALIFICATION IN THE ORIGINAL INVESTMENT ECONOMY

n	Basis	Ω_h upper	Regret upper	Max. dual gap
8	ReLU	5.39764	8.46716	0.000000014
8	Bernstein	5.36565	8.11941	0.000000000
16	ReLU	3.73299	5.28939	0.000000014
16	Bernstein	4.43474	5.81890	0.000000000

Each row is one predeclared, complete two-date construction. Bounds are rounded upward; the exact fractions and dates are retained. Regret upper denotes the proved original-optimum account, not an estimated loss or an attained economic equivalence margin. The maximum dual gap concerns only the declared fixed-feature, fixed-future readout objectives.

The table reports the attained two-date residual account Ω_h, the original-optimum bound $\Omega_h + H_h^\pi$, and the largest exact one-step optimization gap. A smaller qualification objective is not itself an identified expected-cost gain. The matched least-squares ablation in the appendix uses each service's identical features and fixed-future labels; it tests the fitting objective, not a separately propagated controller. Both fitting and uniform qualification are paid, and the explicit state cover is tensor based.

All eight datewise optimization gaps are below 1.4×10^{-8}, against the predeclared tolerance $1/65536$. Their qualified residual bounds are strictly below the matched label-loss bounds; the reductions range from about 18 to 63 percent. The resulting global regret bounds remain conservative, as the table makes explicit. Those objective reductions do not establish a narrow economic equivalence interval or cheaper qualification at the earlier $1/256$ margin.

Every new action value is independently reconstructed using the retained original rational terminal formula or a different exact integration method for the fitted continuation. Continuous-action lower covers, whole-cell choices and primal–dual bounds are then recomputed. These constructions add actual qualified continuations, not new independent cost observations. The stopping theorem has separate exact finite-model regressions; the earlier fixed-sample equivalence intervals are not retroactively given an anytime interpretation.

APPENDIX R: COMPLETE CONSTRUCTED-POLICY COST RECORDS

This appendix reports the entire prospectively specified cost catalogue. The policies are the four acquired actors already returned by the qualification-optimal backward constructions, together with zero investment. The constant action on a cell is read from the original integer action array. No proposed readout is retrained on these paths, and no callback to the optimal Bellman recursion replaces its actual fitted future. At a fixed acquisition resolution, the ReLU and Bernstein constructions have the same nine-coefficient budget and state-and-action information. Cross-resolution comparisons answer a different question and retain their distinct observation contracts.

TABLE R.I: Every marginal original-law cost interval

Observation	Controller	Cost interval	Full width
C	Zero	[0.773560, 0.780120]	0.006559
C	R8	[0.607702, 0.613094]	0.005391
C	B8	[0.684025, 0.690150]	0.006125
C	R16	[0.607008, 0.612371]	0.005362
C	B16	[0.682901, 0.689013]	0.006112
Y	Zero	[0.773561, 0.780120]	0.006559
Y	R8	[0.607716, 0.613107]	0.005390
Y	B8	[0.684034, 0.690159]	0.006125
Y	R16	[0.607022, 0.612384]	0.005361
Y	B16	[0.682910, 0.689022]	0.006112

C is full cash cost and Y integrates the last innovation. Both estimate the same policy cost. All ten intervals belong to the declared joint family. Marginal intervals need not meet the target assigned to paired differences.

TABLE R.II: Every paired policy-cost interval and declared decision

Obs.	First minus second	Difference interval	Sign	Width	Equiv.
C	B8 − B16	[0.000662, 0.001599]	positive	Yes	Yes
C	B8 − R16	[0.076496, 0.078301]	positive	Yes	No
C	R16 − B16	[-0.077164, -0.075372]	negative	Yes	No
C	R8 − B16	[-0.076436, -0.074682]	negative	Yes	No
C	R8 − B8	[-0.077573, -0.075805]	negative	Yes	No
C	R8 − R16	[0.000227, 0.001191]	positive	Yes	Yes
C	Zero − B16	[0.090176, 0.091590]	positive	Yes	No
C	Zero − B8	[0.089052, 0.090453]	positive	Yes	No
C	Zero − R16	[0.166007, 0.168293]	positive	Yes	No
C	Zero − R8	[0.165317, 0.167565]	positive	Yes	No
Y	B8 − B16	[0.000661, 0.001599]	positive	Yes	Yes
Y	B8 − R16	[0.076491, 0.078296]	positive	Yes	No
Y	R16 − B16	[-0.077159, -0.075368]	negative	Yes	No
Y	R8 − B16	[-0.076432, -0.074678]	negative	Yes	No
Y	R8 − B8	[-0.077568, -0.075801]	negative	Yes	No
Y	R8 − R16	[0.000227, 0.001191]	positive	Yes	Yes
Y	Zero − B16	[0.090168, 0.091580]	positive	Yes	No
Y	Zero − B8	[0.089044, 0.090443]	positive	Yes	No
Y	Zero − R16	[0.165995, 0.168280]	positive	Yes	No
Y	Zero − R8	[0.165305, 0.167552]	positive	Yes	No

Width and Equiv. are exact-endpoint decisions at the separate predeclared thresholds, each 1/256. Positive zero-minus-controller means a gain from replacing the zero incumbent. Common-path dependence is retained; the twenty intervals are not twenty independent samples.

The width and equivalence columns are different decisions. A precisely positive gain over the installed zero-investment controller fails equivalence because it identifies a material improvement, not because its computation failed. Conversely, a small point estimate with an interval spanning the chosen margin cannot justify equivalence. All decisions use unrounded rational endpoints. The printed intervals are rounded outwards, so a reader need not infer a sign from a decimal that understates uncertainty.

R.1. *Acquisition, bin integration and arithmetic*

Each path row contains two initial-state and two innovation bin indices. A forty-bit bin represents its full original-law continuous cell. The implementation does not replace the conditional distribution within the bin by a midpoint. At each date, a rectangle enclosing the true state can intersect several acquired cells. The selected action range then includes every cell in that rectangle. Closed upper boundaries are included conservatively. The endpoint calculation propagates that entire range through the stage cost and next transition.

A zero action and every stored constructed action are feasible on the full cell. In particular, the action quantum is checked against the minimum state-dependent capacity on the cell before costs are evaluated. State uncertainty is therefore not silently removed by using the lower-cell center as the true state. Acquisition ambiguity, continuous-bin variation and arithmetic expansion all contribute to the retained numerical width. They can interact, so a decomposition into three separately additive causal effects is not inferred from a single interval run. The exact aggregate numerical term is reported alongside independent ambiguity counts and the two arithmetic implementations' endpoint differences.

TABLE R.III: Exact sources of paired interval width

Obs.	Contrast	Numerical	Variance terms	Support term
C	B8 − B16	0.000000001	0.000020	0.000917
C	B8 − R16	0.000000001	0.000888	0.000917

Obs.	Contrast	Numerical	Variance terms	Support term
C	R16 − B16	0.000000001	0.000874	0.000917
C	R8 − B16	0.000000001	0.000837	0.000917
C	R8 − B8	0.000000001	0.000850	0.000917
C	R8 − R16	0.000000001	0.000046	0.000917
C	Zero − B16	0.000000001	0.000497	0.000917
C	Zero − B8	0.000000001	0.000483	0.000917
C	Zero − R16	0.000000001	0.001368	0.000917
C	Zero − R8	0.000000001	0.001331	0.000917
Y	B8 − B16	0.000000001	0.000020	0.000917
Y	B8 − R16	0.000000001	0.000887	0.000917
Y	R16 − B16	0.000000001	0.000874	0.000917
Y	R8 − B16	0.000000001	0.000837	0.000917
Y	R8 − B8	0.000000001	0.000850	0.000917
Y	R8 − R16	0.000000001	0.000046	0.000917
Y	Zero − B16	0.000000001	0.000495	0.000917
Y	Zero − B8	0.000000001	0.000482	0.000917
Y	Zero − R16	0.000000001	0.001367	0.000917
Y	Zero − R8	0.000000001	0.001330	0.000917

Unrounded components sum exactly to the recorded width. The numerical term is the mean retained full-bin endpoint width, including acquisition and arithmetic effects; these are not asserted to be separately additive causes. Variance terms use exact endpoint sample variances and upward rational square roots.

The first evaluator uses centered polynomial expressions with binary64 outward operations. The second expands the cost and transition polynomials and uses integer floor and ceiling on a 2^{-48} grid. It obtains acquired-action extrema by directly enumerating each distinct intersected rectangle rather than reading the first evaluator's range tables. Thus it shares neither floating arithmetic nor cost, transition or lookup routines with the production evaluator. The stored endpoints use the hull, not the more optimistic intersection, of these two evaluations. Intersection with the global primitive support is justified separately.

Every new full-bin path is recomputed by both implementations during replay. A further exact-rational midpoint calculation checks 512 predetermined rows for each policy, including both estimators. Synthetic boundary tests exercise bins at the state and innovation extremes and across acquired-cell boundaries. These pointwise tests supplement the interval-propagation proof; they are not called full-bin verification by themselves. The historical aliasing counterexample and its corrected R67 execution remain unchanged.

R.2. *Statistical allocation*

There are 30 estimands and 60 one-sided endpoint events in one fixed family. The empirical-Bernstein radii use the common log allowance ten. The union bound is at most $120e^{-10} < 1/100$, verified independently by a rational lower partial sum for e^{10}. All endpoint sums and sums of squares are integer calculations, so no floating sample-variance cancellation enters the confidence endpoints. Upward rational square roots enlarge rather than shrink the radii.

The new family is distinct from the earlier R69 optimal-advantage-centered inference family. Combining conclusions from different families requires adding their error allowances. The R73 construction records contain no new independent cost rows, and replay of the R77 rows adds none. The sample stream, row count, bin precision, policy hashes, estimands and margin are fixed before execution. A failure of the declared width target is retained; it does not authorize more samples after inspecting the signs. No optional-stopping interpretation is assigned to this fixed-sample experiment.

R.3. *Paid work and adoption*

The parent process. The complete five-controller service took 105.049 seconds from process launch through shutdown and log flushing. Exact statistical postprocessing took 8.204 seconds; process peak resident memory was 189128 KiB. Each arithmetic implementation evaluated 2621440 complete policy trajectories, and the additional rational midpoint check covered 2560 predetermined policy trajectories.

TABLE R.IV

MEASURED CONSTRUCTION AND COMPLETE INDEPENDENT EVALUATION PHASES

Policy	Prior build (s)	Float (s)	Integer (s)	Through output (s)	Ambiguities
Zero	–	1.890	14.528	18.184	0
R8	1.174	1.747	15.673	19.407	0
B8	3.174	1.729	14.998	18.581	0
R16	4.793	1.692	15.463	19.353	0
B16	10.617	1.690	14.931	18.656	0

Prior construction clocks are inherited unchanged and include the original matched fitting ablations. Through-output time includes lookup preparation, both evaluators, exact midpoint checks and compressed arrays. The two evaluation columns are only components of that total. Ambiguities count endpoint rectangles spanning more than one actor cell. These are inference-service timings, not new generator cold-start competitions.

The recorded number of original-optimum value queries is zero. CPU frequency is observed where available, not claimed locked.

The acquired-action rectangle tables are paid preprocessing, not a free neural operation. Their simple exhaustive construction is practical at the declared small covers but is not claimed to scale linearly in state dimension. The independent integer implementation deliberately pays for another complete calculation. The parent service also pays for reading and hashing controller records, both evaluations, exact midpoint checks, statistical inference, compression, durable output and shutdown. These costs qualify the returned policies; they do not retime the original backward construction services or establish a new cold-start generator ranking.

Corollary 14 provides a priced decision for every declared deployment count and monetary scale. In particular, the gain interval in Table I yields an entire replacement-fee frontier, rather than a favorable isolated fee. The zero-investment incumbent is a feasible installed policy in the original economy. An improvement over it does not establish dominance over an adaptive Bellman search or over every conventional basis. The matched feature comparison and the retained conventional-query catalogue must be read separately.

The experiment's units are those of the theoretical investment loss. A currency calibration, an empirically estimated installation fee and a deployment-frequency model would be additional economic assumptions. The mathematical adoption interval states their role explicitly and remains valid for all such nonnegative prices on the same coverage event. It therefore supports a conditional purchase calculation without manufacturing an empirical welfare estimate.

APPENDIX S: SIMULTANEOUS CHOICE AMONG CONSTRUCTED CONTROLLERS

The preceding experiment qualifies the same complete controllers returned by the NBO fits. A decision maker must also choose between them. Marginal upper costs give the valid rule in Corollary 14, but discard the more precise paired observations. We retain all available comparisons on their original simultaneous event. No additional simulation, independence assumption between estimators, or allocation of statistical error is required.

Let J_i be the expected original-law cost of constructed controller i, for $i = 1, \dots, q$. Introduce an anchor $\theta_0 = 0$. Every reported marginal interval $[L_i, U_i]$ supplies constraints $\theta_i \le U_i$ and $-\theta_i \le -L_i$. Every reported paired interval $[l_{ij}, u_{ij}]$ supplies

$$\theta_i - \theta_j \le u_{ij}, \qquad \theta_j - \theta_i \le -l_{ij}.$$

Use both estimators when their intervals belong to the joint family of Theorem 13. Denote the resulting bounded set of cost vectors by $\mathcal{C}$. An edge $a \to b$ of weight c represents $\theta_b - \theta_a \le c$, and $d(a, b)$ is its shortest-path distance. These are classical difference constraints; the computational reduction is not a new graph algorithm (Floyd, 1962). Its role here is to turn paid NBO cost observations into an attainable, price-uniform choice certificate.

THEOREM 12—Sharp catalogue choice: *Suppose all marginal intervals are finite and $\mathcal{C} \neq \varnothing$. For every i, j, the tight upper bound implied by the retained inequalities is*

$$\max_{\theta \in \mathcal{C}} (\theta_i - \theta_j) = d(j, i). \tag{17}$$

For any declared resource charges $w_1, \dots, w_q \geq 0$, put

$$\Gamma_i(w) = \max_{1 \leq j \leq q} \{d(j,i) + w_i - w_j\}. \tag{18}$$

Then $\Gamma_i(w)$ is exactly the largest possible cost regret of controller i over $\mathcal{C}$. A minimizer $\widehat{i}$ of $\Gamma_i(w)$ minimizes that worst-case regret among the constructed controllers. On the original simultaneous coverage event,

$$J_{\widehat{i}} + w_{\widehat{i}} - \min_j (J_j + w_j) \leq \min_i \Gamma_i(w).$$

The inequality holds for all charges on that same event, including charges chosen after inspecting the intervals. Controller i is optimal for every cost vector in $\mathcal{C}$ exactly when $\Gamma_i(w) = 0$.

The proof in the Numerical and Development Appendix constructs a feasible cost vector attaining each bound. Sharpness refers to the retained interval inequalities, not to all information in the raw paths or to a statistically optimal procedure. The comparator is the qualified finite catalogue, not the original continuous Bellman optimum; the earlier original-optimum guarantees remain necessary. A negative constraint cycle means inconsistent inputs and stops selection rather than authorizing a silently repaired interval. Closure uses $O(q^3)$ rational additions and comparisons and $O(q^2)$ storage; rational bit complexity and all prior qualification work remain additional costs.

An identified ordering and a priced refinement. In the fixed constructed-policy catalogue, the joint intervals identify

$$J_{\text{ReLU,16}} < J_{\text{ReLU,8}} < J_{\text{Bernstein,16}} < J_{\text{Bernstein,8}} < J_{\text{zero}}.$$

Thus the finest ReLU controller is the unique zero-charge cost minimizer in this catalogue. At matched resolutions, its per-deployment gain over Bernstein is enclosed by $[0.0758058, 0.0775680]$ for $n = 8$ and $[0.0753723, 0.0771589]$ for $n = 16$, rounded outward. These compare actual attained continuations and their complete feasible actors, not optional query proposals. They do not reverse the adverse cold-start and reuse findings for the different recursive-query controllers.

Let $[l,u]$ be the closed interval for $J_{\mathrm{ReLU},8} - J_{\mathrm{ReLU},16}$. Suppose only the finer controller bears an incremental per-deployment charge $f \geq 0$, all other charges being equal. The other three policies are already dominated by the coarser ReLU controller. The two remaining regret certificates are therefore

$$\Gamma_{16}(f) = (f-l)_+, \qquad \Gamma_8(f) = (u-f)_+.$$

The fine controller is universally optimal for $f \leq l$; the coarse one is universally optimal for $f \geq u$. Between those thresholds, the regret-minimizing choice changes at $(l+u)/2$. Here $[l,u] \subset [0.0002277, 0.0011901]$ and the exact switch is approximately 0.00070894. Exact endpoints, all adjacent gaps and attaining cost vectors are deposited. Transitive closure does not further tighten the paired bounds in this particular catalogue; no such improvement is claimed.

For m deployments and monetary scale $v > 0$, replace f by the incremental total charge divided by mv. This gives a complete conditional fee region without assigning a monetary value to normalized losses or inferring a deployment horizon from the sample count. Acquiring finer information is an economic choice only after its actual incremental costs are specified.

APPENDIX T: PROOF AND IMPLEMENTATION OF SIMULTANEOUS CONTROLLER CHOICE

PROOF OF THEOREM 12: If $\theta \in \mathcal{C}$, summing its constraints along a path from j to i gives $\theta_i - \theta_j$ at most the path's length. Hence it is at most $d(j,i)$. Feasibility rules out a negative cycle, because summing the constraints around a cycle has left side zero. Every vertex has a finite path to and from the anchor, by the finite marginal bounds. All shortest distances are therefore finite and $d(j,j) = 0$.

Fix j and define a cost vector, including the anchor, by

$$\theta_k^{(j)} = d(j,k) - d(j,0), \qquad k = 0, \ldots, q.$$

For each edge $a \to b$ of weight c, the shortest-path triangle inequality gives $d(j,b) \leq d(j,a) + c$. Thus $\theta_b^{(j)} - \theta_a^{(j)} \leq c$, and $\theta_0^{(j)} = 0$. This vector belongs to $\mathcal{C}$ and attains $\theta_i^{(j)} - \theta_j^{(j)} = d(j,i)$. This proves (17). The same construction proves that a graph with

no negative cycle and finite anchor paths has a feasible cost vector; hence the cycle check is an exact consistency test.

For a fixed controller i, its cost regret at θ is

$$\theta_i + w_i - \min_j(\theta_j + w_j) = \max_j\{\theta_i - \theta_j + w_i - w_j\}.$$

The maximum is finite, so taking its supremum over $\mathcal{C}$ commutes with the maximum over j. The preceding attainment result gives exactly (18). The term $j = i$ is zero, so $\Gamma_i(w) \geq 0$. For a maximizing j, the vector $\theta^{(j)}$ attains this regret. Minimizing over i proves the claimed deterministic minimax property. On the coverage event the true cost vector belongs to $\mathcal{C}$; inserting it proves the statistical conclusion without another probability bound. All subsequent arguments are deterministic for every w, which establishes price-uniformity. Finally, nonnegative regret is identically zero exactly when its maximum is zero. *Q.E.D.*

The ordinary Floyd recurrence evaluates the all-pairs distances with $(q+1)^3$ candidate relaxations. The implementation uses exact fractions, including the rational confidence endpoints, rather than shortest paths through rounded printed entries. Its independent cross-check runs a separate single-source Bellman–Ford recurrence from every vertex. Each returned attaining vector is checked against every original edge and against every claimed attained difference. Negative self-loops and longer negative cycles are regression cases; an inconsistent graph cannot pass by resetting its diagonal.

T.1. *Refinement charges and the finite comparison*

Let the fine and coarse ReLU controllers be denoted by A and B, with $J_B - J_A \in [l, u]$, where these endpoints are the tight implications of $\mathcal{C}$. Suppose all policies except A have a common charge and A has an additional charge f. The retained inequalities identify B as less costly than every other controller in the catalogue, so those controllers cannot be a best-cost comparator or a regret-minimizing choice. Equation (18) reduces to

$$\Gamma_A(f) = \max\{0, f - l\}, \qquad \Gamma_B(f) = \max\{0, u - f\}.$$

For $f \leq l$, only A can have positive strict cost advantages against all alternatives; its worst-case regret is zero. For $f \geq u$, B has zero worst-case regret. On $[l, u]$, the difference of the two regrets is $2f - l - u$, proving the switching formula. At a boundary, universal weak optimality, rather than strict optimality, is the appropriate statement. The independently checked exact fee probes include both boundaries, the switching point, and points outside the ambiguity interval.

TABLE T.I

ADJACENT GAPS IN THE CONSTRUCTED-POLICY COST ORDERING

Higher minus lower cost	Lower	Upper
ReLU, $n = 8$ minus ReLU, $n = 16$	0.0002277	0.0011901
Bernstein, $n = 16$ minus ReLU, $n = 8$	0.0746820	0.0764312
Bernstein, $n = 8$ minus Bernstein, $n = 16$	0.0006622	0.0015984
Installed zero minus Bernstein, $n = 8$	0.0890522	0.0904430

All endpoints are rounded outward. Both original estimators and all marginal and paired intervals enter one consistent difference system. These are the same frozen actors and the same 524,288 rows, not a new experiment. Charges are set to zero only for this statistical cost ordering.

TABLE T.II

PER-DEPLOYMENT GAINS AVAILABLE TO PAY INCREMENTAL CHARGES

Replacement	Lower	Upper
ReLU versus Bernstein, $n = 8$	0.0758058	0.0775680
ReLU versus Bernstein, $n = 16$	0.0753723	0.0771589
Refinement from ReLU $n = 8$ to $n = 16$	0.0002277	0.0011901

A positive lower gain licenses any smaller incremental total charge in normalized cost units. For m deployments and monetary scale v, multiply the endpoints by mv. Construction, acquisition, qualification and installation charges remain explicit; seconds are not assigned an unobserved monetary price.

Both original cash and conditional-estimator intervals qualify the same expected costs and are covered by the same thirty-estimand event. Intersecting their implica-

tions therefore needs no independence between estimators and no new multiplicity allowance. It does not convert two estimators on one stream into two independent datasets. The exact constraint ledger, closure, attaining cost vectors, zero-charge regrets and fee probes are in the machine-readable decision record.

All ten unordered contrasts and five marginal costs from each estimator enter the computation. In this particular dataset, transitive paths do not improve the direct paired endpoint intersections. The new decision result is consequently not presented as a new precision experiment. It uses the already purchased joint information to establish an identified catalogue ordering and sharp conditional choice regions. A separate exact test provides a graph in which transitivity strictly tightens an indirect comparison; that test is a mathematical regression, not another observation from the investment economy.

The qualification process is part of the cost of learning these regions. Its historical complete service clock is retained, not replaced by the much shorter graph-processing clock. Controller construction, acquisition, qualification, deployment and installation charges are distinct. A common sunk cost cancels from comparisons only under a decision problem in which it is indeed common and sunk. The arbitrary-charge theorem does not authorize dropping a method-specific cost. Likewise, the catalogue includes the actually attained ReLU and Bernstein actors; it does not include the different recursive adaptive-search controller merely because that controller was evaluated in an earlier experiment.

APPENDIX U: PROOF, INSTALLATION COSTS, AND FINITE DEPLOYMENT COHORTS

PROOF OF THEOREM 15: For fixed j, each feasible cost vector satisfies $\theta_i - \theta_j \leq d(j,i)$. Nonnegative allocation weights therefore give

$$\sum_i p_i(\theta_i - \theta_j) \leq \sum_i p_i d(j,i).$$

The potential $\theta_i^{(j)} = d(j,i) - d(j,0)$ is feasible and attains all of these differences at once. Hence equality is attained for every weighted sum in this display, not just for each pair separately. Commuting the maximum over $\mathcal{C}$ with the finite maximum

over j gives (39); a maximizing row supplies an attaining potential. The argument would not hold for arbitrary separately sharp pairwise bounds lacking a common attaining vector.

Minimizing the largest row payoff is the program

$$\min_{p,z} z \quad \text{subject to } p \geq 0, \quad \mathbf{1}^\top p = 1, \quad Hp \leq z\mathbf{1}.$$

Its dual maximizes y subject to $\lambda \geq 0$, $\mathbf{1}^\top \lambda = 1$ and $H^\top \lambda \geq y\mathbf{1}$. Both simplexes are compact and the matrix is finite, so linear-programming duality gives equal attained values. Independently of optimality, every feasible pair satisfies

$$\min_i (H^\top \lambda)_i \leq \lambda^\top H p \leq \max_j (Hp)_j.$$

Taking the appropriate infimum or supremum proves the computable gap bound. For rational input, exact rational vertices exist, and the certificate only requires feasibility and these inequalities, not a floating-point solver status. Finally, on the original coverage event the true cost vector belongs to $\mathcal{C}$. All preceding inequalities are deterministic for every p, w on that event. The allocation draw is independent of the deployment trajectory and takes place before information acquisition, so its expected cost is $p^\top J$. This proves the stated economic interpretation. It is not a pathwise guarantee for a particular random assignment. *Q.E.D.*

U.1. *Installed subsets rather than free simultaneous availability*

Suppose a decision maker serves $m \geq 1$ deployments and may install a nonempty subset S at total normalized cost $K(S)$. Write $K_i = K(\{i\})$, and assume $K(S) \geq K_i$ for every $i \in S$. The benchmark installs the best single controller, paying its own K_j. For p supported on S, the exact worst-case average regret is

$$\mathcal{R}_S(p, w) = \max_j \left\{ \sum_i p_i H_{ji}(w) + \frac{K(S) - K_j}{m} \right\}. \tag{19}$$

Indeed, the installation term is independent of θ, and the same attaining potential proves the formula. The assumption on K makes the cost of an allocation at least the weighted average of its corresponding singleton costs, and hence at least their

minimum; the regret is nonnegative. For each subset, (19) is a linear program on Δ_S. Enumerating the nonempty subsets and retaining the least optimum yields the exact installation-and-allocation decision. It requires at most $2^q - 1$ programs and is not advertised as dimension-free optimization. Lower bounds from the dual of every subset, together with an attaining primal allocation for the chosen subset, certify a global solution. A controller outside the installed subset remains a possible oracle comparator and must not be removed from the rows.

U.2. *Two-controller formula and extra setup threshold*

With only the fine controller charged f, put $a = f - l$ and $b = u - f$. Within the ambiguity region both are positive and $a + b = u - l$. If p is the fine share, the two worst-case comparator regrets are pa and $(1 - p)b$. Their maximum is minimized at their intersection, giving $p = b/(a + b)$ and value $ab/(a + b)$. Outside that region a uniformly optimal pure controller has zero regret. If $l = u$, the gain is known exactly: use the fine controller below l, the coarse one above it, and any allocation at equality; no division by a zero width is needed.

When both controllers incur additional cost κ only for $0 < p < 1$, their optimal nontrivial allocation is unchanged and its value is $ab/(a + b) + \kappa$. The best pure value is $\min\{a, b\}$. Subtracting gives the exact saving $\min\{a, b\}^2/(a + b)$ and proves (41). Equality gives a tie. A positive extra cost can make the pure rule optimal; the deployment analysis never assumes that loading or operating an additional controller has zero cost. Formula (19) handles unequal singleton installation charges as well.

U.3. *A deterministic allocation to a finite cohort*

Random assignment is unnecessary when m homogeneous deployments with the same expected cost law can be assigned in advance. Deploy A on exactly k tasks and B on the others, without choosing tasks according to unqualified outcome information. The average expected cost is the mixture with $p = k/m$; independence between tasks is not needed for this expectation identity. For $l < f < u$, a minimizing integer k is one of $\lfloor mp^*_A \rfloor$ and $\lceil mp^*_A \rceil$. This follows because the piecewise-linear

regret decreases up to p_A^* and increases afterwards. Nearest rounding also supplies

$$0 \le V_m(f) - V(f) \le \frac{\max\{f-l, u-f\}}{2m}.$$

An extra two-controller setup charge is included only for $0 < k < m$; both pure endpoints must then also be tested. Nonhomogeneous deployment laws require separately qualified cost vectors and are not covered by relabeling the existing sample as a cohort.

TABLE U.I

QUALIFIED ACQUISITION-FEE CHOICES AND DEPLOYMENT ALLOCATIONS

Fee position	Fine share	Pure regret upper	Mixed regret upper	Setup saving lower
0	1	0.00000000	0.00000000	0.00000000
1/4	3/4	0.00024058	0.00018044	0.00006014
1/2	1/2	0.00048115	0.00024058	0.00024057
3/4	1/4	0.00024058	0.00018044	0.00006014
1	0	0.00000000	0.00000000	0.00000000

Fee position t means $f = l + t(u-l)$. Both controllers were qualified under the original law; a share selects an entire controller before information acquisition. Regret is worst-case expected loss relative to the best single catalogue controller. The last column is the largest extra normalized cost of maintaining both controllers compatible with strict improvement; equality gives a tie. Values are outward rounded. No row is a new simulation or a calibrated installation fee.

The deposited implementation checks exact probability feasibility, all original cost inequalities at each attaining potential, and equality between primal and dual payoffs. Two-policy formulas are checked against a separate exhaustive rational vertex solver on the full five-controller catalogue. Finite-cohort choices are checked against every integer allocation. The original thirty cost estimands, 524,288 evaluation rows, fitted coefficients, scientific clocks and adverse selector comparisons remain unchanged. The optimization tables contain no new simulation or timing evidence. The sharpness claim is relative to the retained interval system; neither an exact decision solution nor a positive finite comparison establishes a population-wide neural ranking.

APPENDIX V: NEURAL BELLMAN QUERIES WITHOUT A STATE LATTICE

The preceding construction makes numerical error accountable, but a complete tensor reference can absorb the benefit of a fitted continuation. We therefore change the computational organization, not the economic problem. A Neural Bellman selector proposes an action at the state actually acquired. A recursive interval query then bounds the original Bellman value at that state and checks the proposed action against the entire continuous feasible interval. No conventional policy is precomputed at all states. The controller is the resulting callable algorithm, not the unverified proposal alone.

The distinction matters for attribution. Learning may reduce the work needed to locate a useful action. It does not supply the inequalities used to certify that action. The adaptive conventional comparator receives the same inequalities and the same original-optimum target. The trained value-fit policies in the preceding study remain distinct from the action-query predictor introduced here.

V.1. *A curvature lower bound on an adaptive action partition*

Use the original scalar economy of Section 4, with r dates remaining. Write V_r for its optimal value and $Q_r(x,a)$ for the value of a first action followed by the optimal future. Let G_r, M_r be the primitive constants in Theorem 4, indexed by remaining horizon, and put

$$H_r = \frac{21}{4} + \beta M_{r-1}\frac{5d}{32}, \qquad \Lambda_r = \frac{13}{16} + \beta G_{r-1}\frac{3d}{8}.$$

Thus H_r is an upper curvature bound in the scalar action, and Λ_r bounds the absolute action derivative. Neither depends on a fitted model.

LEMMA 13—Parabolic minorants: *Suppose f is H-semiconcave on $[l,u]$, and $L_l \le f(l)$, $L_u \le f(u)$. For $s \in [0,1]$,*

$$f(l + s(u-l)) \ge (1-s)L_l + sL_u - \frac{H}{2}(u-l)^2 s(1-s) =: p(s).$$

Consequently $\min_{[0,1]} p$ is a lower bound for the minimum of f on the interval. With $K = H(u-l)^2/2$ and $D = L_u - L_l$, this lower bound is L_l if $D \ge K$, L_u if $D \le -K$, and

otherwise

$$\frac{L_l + L_u}{2} - \frac{K}{4} - \frac{D^2}{4K}.$$

The endpoints L_l, L_u need not themselves belong to a convex interpolant. Increasing K preserves the lower-bound property.

The proof is in Supplement, Section 2.6.

At an acquired state, retain an ordered partition of the feasible action interval and an outward value bracket at every partition point. The minimum of the parabolic minorants bounds the continuous-action infimum from below. The smallest upper endpoint supplies an actual feasible witness. A neural or polynomial proposal inserts another point into this partition; it cannot remove the endpoints or any part of the feasible interval. The running lower bound is the maximum of all previously valid global lower bounds, so new information cannot weaken the certificate. The upper bound is the minimum over every evaluated feasible witness.

V.2. *Recursive queries under the original continuous law*

For an endpoint action with $r > 1$, divide the scalar innovation into q_r equiprobable bins and query V_{r-1} at each conditional-mean next state. If each child returns width at most η_r, the endpoint bracket has width at most

$$\beta\eta_r + \frac{\beta M_{r-1} d}{6144 q_r^2} + \omega_r. \tag{20}$$

Here ω_r includes outward primitive arithmetic and the Lipschitz expansion from an interval-valued next state to its queried center. Convexity provides the lower conditional-mean inequality; semiconcavity and the exact conditional variance provide the upper allowance. The innovation remains continuous. There is no state-interpolation term because a child value is queried rather than read from a state lattice.

For a requested local gap δ, the implementation sets $\eta_r \le \delta/(4\beta)$ and chooses a dyadic q_r so that the second term in (20) is at most $\delta/4$. A separate endpoint-width guard charges arithmetic and the small capacity-rounding remainder. Failure of

that guard is a precision failure, not a successful return. The terminal query uses analytic integration instead of recursive quadrature.

THEOREM 14—A state-grid-free NBO service: *Consider the original scalar investment economy. At every recursive query, suppose the outward endpoint and capacity-transfer accounts give width at most $5\delta/8$. Let the lower bound be the running maximum described above. Refine an interval attaining the smallest current minorant whenever the global upper-minus-lower gap exceeds δ. Each new point must lie between one fifth and four fifths of that interval. Arbitrary additional learned proposals may be inserted initially. Then:*

(i) every returned interval $[L_r(x), U_r(x)]$ contains $V_r(x)$, and its returned feasible action $a_r(x)$ satisfies

$$Q_r(x, a_r(x)) - V_r(x) \le U_r(x) - L_r(x) \le \delta;$$

(ii) the real-arithmetic refinement terminates. With at most one interior learned proposal, a sufficient point budget is

$$6 + \left\lceil 5c(x)\sqrt{H_r/(3\delta)} \right\rceil; \tag{21}$$

(iii) for the policy that calls this service at every subsequent date, the original-law loss satisfies

$$J_T^{\pi}(x) - V_T(x) \le \sum_{t=0}^{T-1} \beta^t \left[\delta_t + \left(8 + \frac{11\beta d G_{T-t-1}}{16} + d G_{T-t} \right) 2^{-b} \right], \tag{22}$$

when the state is acquired downward at b bits per coordinate and every returned action is feasible at that acquired state. All action rounding must precede its endpoint evaluation. The inequalities hold uniformly in the true initial state, not only at the states used in an experiment.

The proof is in Supplement, Section 2.7.

The finite-precision contract is explicit. The executed service uses the width guard, a 512-point cap, and a 32-bit downward action lattice. A precision or cap failure withholds the floating certificate and invokes a separately counted exact-rational enumeration. This fallback constructs values only at the required descendants; it does not load a precomputed conventional policy. The rational fallback chooses

dyadic action and quadrature counts until each cover or quadrature allowance is at most one quarter of the request, and requests child width at most one quarter divided by the discount. Provided $\delta > 8\Lambda_r 2^{-32}$ at every recursive level, its exact brackets have width at most three quarters of the request. This proves a finite return for every acquired state in the declared tasks, including states absent from the workload. Rational bit-operation cost is additional work and is not replaced by a unit-cost floating clock. The mathematical point budget is checked against the declared tolerance range; termination in an arbitrary floating implementation is not inferred merely from the real-arithmetic argument. Fitting convergence is unnecessary for validity: any finite prediction, and even its clipped replacement when numerically invalid, is only a proposed query.

V.3. *A certified analytic terminal kernel*

The last decision admits an especially strong common comparator. Let $y = F(x,0,0)$, $b_0 = 1/2 - 2\,\mathrm{mean}(y)$, and $\gamma_j = B_j$. Define

$$D(x) = \frac{8}{d}\sum_j \gamma_j (y_j - 5/8) + \frac{1}{2d}\sum_j (\gamma_j - \gamma_{j+1})(y_j - y_{j+1}).$$

For every even d, the exact terminal action derivative is

$$16a^3 + \left(2 + \frac{41\beta}{32}\right) a + \beta D(x) - 3\beta(b_0 - 3a/4)_+. \tag{23}$$

Each branch is a strictly increasing cubic. Its floating closed-form root supplies only a candidate. The independently evaluated outward derivative and exact-law objective certify that candidate; a safeguarded derivative bracket remains available if the candidate is inaccurate.

For completeness, if f is μ-strongly convex and an outward interval encloses $f'(a)$, the largest compatible projected derivative residual R gives

$$f(a) - \min_{[0,c]} f \le R^2/(2\mu).$$

At zero, a nonnegative derivative has residual zero; at capacity, a nonpositive derivative has residual zero. Otherwise the absolute derivative bound is sufficient.

This follows by minimizing the strong-convexity supporting quadratic over the feasible interval and, conservatively, over the line. Here $\mu = 2$. Equation (23), its $41/32$ and $9/4$ coefficients, analytic shock moments, and outward derivatives are independently checked against rational formulas. Every method uses this kernel; it is not credited to the neural predictor.

V.4. *Work, prediction, and the economic object*

Let $A_r(\delta)$ be a bound on action endpoints at a query. Excluding training, a scalar operation account obeys

$$W_1(\delta) = O(dA_1), \qquad W_r(\delta) \le CdA_r(\delta) + A_r(\delta)q_r W_{r-1}(\delta/(4\beta)).$$

The constants dG_r, dM_r are bounded independently of d, so the declared endpoint and quadrature budgets need not grow exponentially with the state dimension. Primitive evaluations still cost $O(d)$. This removes the state lattice, not the horizon or shock-tree cost. The executed vectorization batches a query's innovation descendants. Its largest live state batch is reported; it does not claim the smaller memory of a fully streamed implementation. Increasing the horizon or the shock dimension may still be expensive. The new execution is scalar-control; the earlier full two-control result is retained rather than relabeled as a query-service experiment.

A useful prediction can reduce the endpoint count or return its own action from the initial partition. An inaccurate prediction can instead add work. This is a prediction-dependent algorithm, not a theorem of neural dominance. Learned warm starts have their own literature, including Sakaue and Oki (2023); their discrete-convex setting differs from the original-law Bellman intervals here. Convex control policies and learned optimization parameters are studied by Agrawal, Barratt, Boyd, and Stellato (2020). Our comparison identifies the extra proposal cost and the resulting query transcript rather than claiming priority for learning a warm start.

The purchase is now a complete callable controller with a declared error contract. Its training cost, every subsequent certified query, and its return-time cost belong to the same service. A reduction in queried state storage is economically relevant for repeated decisions only after these costs are counted. The original cost normalization

is maintained. Workload seconds and normalized expected-cost loss are reported separately; a price per unit of computation can convert the former to the latter only as an explicitly specified decision-maker parameter.

APPENDIX W: OPERATIONAL PREDICTION CERTIFICATES AND READOUT TRAINING

Proposition 5 separates witness accuracy from lower-certificate localization. Its distance to an unknown minimizer is useful for analysis but is not a stopping observable. We now replace that distance by an upper envelope computed from information already purchased. This gives a certificate before an additional Bellman query, a training criterion measured in unresolved verification work, and a constructive neural readout with a computable optimization gap.

W.1. *A decision before the next query*

Fix a state, a date, and the current action partition. Write $f(a) = Q_t^*(x,a)$, and suppose f is μ-strongly convex on the feasible interval. Let $\overline{f}_l, \overline{f}_u$ be verified upper endpoints at two consecutive actions $l < u$. Let L be the current lower bound for the original feasible infimum, including the capacity and arithmetic allowances. For $a = l + s(u-l)$, define

$$C(a) = (1-s)\overline{f}_l + s\overline{f}_u - \frac{\mu}{2}(a-l)(u-a), \qquad 0 \le s \le 1. \tag{24}$$

Every symbol in this expression is either a primitive bound or an existing endpoint. No derivative or action at the unknown optimum occurs.

THEOREM 15—Pre-query closure: *Under the preceding conditions, $f(a) \le C(a)$ throughout $[l,u]$. Let $a_\Delta \in [l,u]$ be the actual feasible rounded proposal. An outward upper endpoint $\overline{C}(a_\Delta)$ satisfying*

$$\overline{C}(a_\Delta) - L \le \delta \tag{25}$$

certifies $f(a_\Delta) - \inf_{a \in A(x)} f(a) \le \delta$ without another Bellman query. If the screen does not close, the protected refinement of Theorem 4 retains its validity and finite query bound. The complete acquired policy therefore retains the original-optimum account of Theorem 14.

Equivalently, the union over partition intervals of the sublevel sets $\{a : C(a) - L \le \delta\}$ is a computable certified action set. Each constituent is a quadratic sublevel interval intersected with its partition cell. Endpoints obtained numerically must be rounded inward, or the final action must be checked by (25).

Proof. See Supplement, Section 3.1.

The screen is shared by every comparator. In particular, minimizing the quadratic upper envelope is an available conventional proposal. Prediction is valuable only to the extent that its proposed location or evaluation cost improves the observed service. The executed neural rule screens its feasible prediction on the partition interval containing it, even when that location is outside the protected part of the currently critical interval. A failed screen does not authorize an unprotected split.

W.2. *A computable loss for unresolved work*

A *certificate context* z consists of the state, remaining horizon, tolerance, partition endpoints and their verified bounds. A proposal $a_\theta(z)$ is feasible and is computed without another Bellman query. Define its verified score and a margin loss by

$$s_\theta(z) = \overline{C}_z(a_\theta(z)) - L_z, \tag{26}$$

$$\ell_\tau(z,\theta) = \min\left\{1, \frac{(s_\theta(z) - \delta_z + \tau)_+^2}{\tau^2}\right\}, \qquad \tau > 0. \tag{27}$$

Let $K(z)$ be a valid cap on further endpoint queries if the screen fails and the protected fallback is followed. This is a count of endpoint queries at the current state. The recursive cost of each endpoint is charged through the earlier descendant-work recursion. On contexts satisfying the declared precision contract, the real-arithmetic and rational-recovery bounds supply finite such caps.

THEOREM 16—Certificate loss and additional work: *If $N_\theta(z)$ is the number of additional endpoint queries after the screen, then*

$$N_\theta(z) \le K(z)\mathbf{1}\{s_\theta(z) > \delta_z\} \le K(z)\ell_\tau(z,\theta). \tag{28}$$

Consequently, an achieved margin loss bounds the average additional queries on any specified finite context catalogue, with no unknown minimizing action.

For a distributional assertion, condition on construction of a finite family of M predictors. Suppose $z_1, \dots, z_n$ are independent validation contexts from a declared predictor-independent law, and $0 \le K(z) \le K_{\max}$. With probability at least $1 - \alpha$, simultaneously for that family,

$$\mathbb{E} N_\theta(z) \le \frac{1}{n} \sum_{i=1}^{n} K(z_i) \ell_\tau(z_i, \theta) + K_{\max} \sqrt{\frac{\log(M/\alpha)}{2n}}. \tag{29}$$

The cost of obtaining and validating contexts and of evaluating the predictor is additional work, not part of N_θ.

Proof. See Supplement, Section 3.2.

The context-law requirement matters. A stream of recursively generated contexts can depend on earlier predictions; it is not automatically an independent validation sample for (29). The deterministic inequality (28) remains valid context by context. The recorded action-label mean squared error is not substituted for (27). The new loss measures a certificate's unresolved work rather than resemblance to an approximate action label.

W.3. *A constructive ReLU readout with an optimization certificate*

The work loss can be trained without invoking a successful hidden-layer optimizer. Fix a nonnegative affine–ReLU feature map $\phi(z) \in \mathbb{R}^p$ with $\sum_j \phi_j(z) = 1$. Linear hat functions on a fixed coordinate partition are one exact construction; averages of such maps give multivariate features. A box-constrained readout $w \in [0,1]^p$ proposes

$$a_w(z) = l_z + (u_z - l_z) \phi(z)^\top w.$$

The partition interval and features are fixed before optimizing w. Let $\eta_i \ge 0$ cover action rounding and evaluation of the score, so that the computed score of the actual rounded action is at most $C_i(a_w(z_i)) + \eta_i - L_i$. For example, on lattice-aligned endpoints an action Lipschitz bound gives $\eta_i = \Lambda_i \Delta_a$ for downward rounding.

Define

$$F(w) = \frac{1}{n}\sum_i K_i\big(C_i(a_w(z_i)) + \eta_i - L_i - \delta_i + \tau\big)_+^2 + \frac{\lambda}{2}\|w\|_2^2, \qquad \lambda \geq 0. \tag{30}$$

Unlike an action-label objective, this loss directly upper-bounds the additional-work account. It trains the upper witness against the current global lower certificate.

PROPOSITION 17—Certified readout training: *The function F is convex and continuously differentiable on the box. For $g = \nabla F(w)$, put*

$$D(w) = g^\top w - \sum_{j=1}^{p} \min\{0, g_j\}. \tag{31}$$

Then $0 \leq F(w) - \min_{v\in[0,1]^p} F(v) \leq D(w)$. On the training contexts the average additional endpoint-query count is at most $F(w)/\tau^2$. If $\lambda > 0$, the minimizer w^* is unique and

$$\|w - w^*\|_2 \leq \sqrt{2D(w)/\lambda}.$$

All quantities in these conclusions can be bounded by outward arithmetic; they are exact rational quantities when the contexts, features and iterate are rational.

Proof. See Supplement, Section 3.3.

This is a fixed-feature neural construction, not a convergence assertion for unrestricted nonconvex network fitting. The deposited rational implementation uses finite conditional-gradient steps and reports its achieved loss and dual gap. Its regression examples test the proposition; they are not new investment-policy observations. The preceding R67 service comparison retained its preregistered action-label fits. The new comparison in Section F separately executes this certificate-aware objective, with equal-information Bernstein and label-loss controls; no historical fit is relabeled as its minimizer. Prediction-dependent optimization has an established literature (Sakaue and Oki, 2023, Sambharya, Hall, Amos, and Stellato, 2023); the additional object here is a computable original-law Bellman certificate that can license an action before another query and supplies a directly auditable training criterion.

APPENDIX X: TWO CONTROLS AND TWO CONTINUOUS INNOVATIONS

The absence of a state lattice does not remove action or innovation dimensions. We extend the query implementation to the previously specified two-control economy of Section 4: $a_1, a_2 \geq 0$, $a_1 + a_2 \leq c(x)$, the interaction cost is $a_1 a_2/4$, and the second innovation is an independent common-direction uniform shock. The original scalar model and its experiments are unchanged.

At a queried state, triangulate the feasible action simplex. Let v_0, v_1, v_2 be a triangle's vertices with verified lower endpoints l_0, l_1, l_2. If $\theta_j(a)$ are its barycentric coordinates, define

$$p(a) = \sum_{j=0}^{2} \theta_j(a) l_j - \frac{H}{2} \sum_{j=0}^{2} \theta_j(a) \|v_j - a\|_2^2. \tag{32}$$

This is the quadratic $H\|a\|_2^2/2 + b^\top a + c$ interpolating the lower endpoints. Its approximate minimizer is only a proposed support point. For any point $y \in \mathbb{R}^2$, including a numerically displaced point outside the triangle,

$$p(y) + \min_j \nabla p(y)^\top (v_j - y) \leq \min_{a \in \mathrm{co}(v_0, v_1, v_2)} p(a). \tag{33}$$

Thus an outward supporting-plane calculation, not an uncertified quadratic solver, supplies the lower endpoint.

THEOREM 18—Simplicial Bellman query: *For the two-control primitives, (32) is a lower minorant of the true Bellman objective. Taking the smallest valid triangle lower endpoint and the best feasible-action upper endpoint gives a lower–upper certificate for the full continuous simplex. Include the omitted capacity strip and actual action-rounding allowances. Returning when this gap is at most δ gives the same acquired-policy loss recursion as Theorem 14.*

A failed floating triangulation can be replaced by an exact-rational simplex enumeration. Under the stated positive precision slack, this recovery is finite at every finite horizon. With two innovations divided into q bins each, a valid local quadrature error is

$$\zeta = \frac{5\beta M_{t+1} d}{24576 q^2}. \tag{34}$$

If A_r bounds the number of action queries at a state with r dates remaining, a work account has the form

$$W_r \le C_d A_r + A_r q_r^2 W_{r-1}, \qquad C_d = O(d), \tag{35}$$

with separate charges for triangulation, arithmetic and output. There is no tensor state-table factor, but the descendant tree remains.

Proof. See Supplement, Section 3.4.

The executed algorithm uses longest-edge refinement of the lowest triangle and an outward terminal convexity-gap check. Its floating cap is followed by the stated recovery rule, not by an uncertified returned action. The implementation batches descendants but can allocate a descendant array before slicing it into query batches; a reported maximum query batch is therefore not a bound on total live state memory. We report process memory separately. The two-control execution is a verified extension of the NBO query backend, not evidence for a trained two-output selector.

APPENDIX Y: DUAL VERIFICATION AND POLICY-CLASS TRANSFER

This appendix proves the numerical certificate used in Theorem 10 and states the condition needed to extend its coupling argument beyond two dates. The implemented two-date calculation remains the one in the article. No unexecuted longer-horizon optimization problem is counted as part of that experiment.

Y.1. *A lower bound independent of solver success*

Use the notation of (29). The switch problem can be written as

$$\begin{aligned} \inf_{\theta,u,l,L} \quad & u - l + 2\rho G + \alpha L, \\ \text{subject to} \quad & y_i^+ - \phi_i^\top \theta \le u, \quad l \le y_i^- - \phi_i^\top \theta, \\ & s_j^\top \theta \le L, \quad -s_j^\top \theta \le L, \quad L \ge 0, \\ & c_{Cb} + d_{Cb}^\top \theta \le 0, \quad |\theta_k| \le B_k, \qquad \alpha = 2\rho. \end{aligned}$$

The minimum over (u, l, L) at any feasible θ equals $F(\theta)$. For any feasible point, probability vectors p, q give

$$u - l \geq p^\top y^+ - q^\top y^- + \sum_i (q_i - p_i)\phi_i^\top \theta.$$

For nonnegative ξ_j^+, ξ_j^- with total mass at most α,

$$\alpha L \geq \sum_j (\xi_j^+ - \xi_j^-) s_j^\top \theta.$$

The unused mass is nonnegative and $L \geq 0$. Moreover, $\nu(c_{Cb} + d_{Cb}^\top \theta) \leq 0$ for $\nu \geq 0$. Adding this nonpositive expression to the right side of the preceding lower bound cannot invalidate it. Finally,

$$\inf_{|\theta_k| \leq B_k} z^\top \theta = -\sum_k B_k |z_k|.$$

Combining these inequalities proves (29). It proves weak duality for every rationally feasible tuple, including tuples obtained from an unsuccessful numerical optimizer. Strong duality or an attained numerical optimum is unnecessary for the action-stability conclusion. The lower bound may be uninformative, but it cannot assert a successful certificate without the strict comparison in (27).

The independent verifier reconstructs the derivative rows directly. It computes each expected polynomial feature by integrating its expanded monomials through

$$\mathbb{E} Y^p = \frac{v^{p+1} - u^{p+1}}{(p+1)(v-u)}, \qquad Y \sim \mathrm{Unif}[u, v],$$

and each hinge by the integral of its positive-part antiderivative. This differs from the construction's even-moment expansion. Immediate state costs cancel in an action contrast, leaving $b^2 - a^2 + 4(b^4 - a^4)$ and the discounted expected-feature difference. Every coefficient, primitive and multiplier in this replay is rational. The verifier neither imports the construction's LP routines nor calls its feature-expectation or dual-bound functions.

Y.2. *First-disagreement extension*

Let two finite-horizon policies have a common terminal cost and common transition primitives. Suppose they choose the same action outside measurable sets E_t at each date. Assume a common-innovation realization of the actual kernels is available, so trajectories agree while states and actions agree. Let the tail cash cost from date t lie in $[m_t, M_t]$ under either policy, uniformly over feasible paths, and write $H_t = M_t - m_t$. Put $b_0 = 1$ and $b_t = \prod_{j<t} \beta_j$.

PROPOSITION 19—First-disagreement cost bound: *If $\widehat{\pi}$ is the reference policy and D_t is the event that the coupled policies first choose different actions at date t, then*

$$|J^{\pi} - J^{\widehat{\pi}}| \leq \sum_{t=0}^{T-1} b_t H_t \Pr(D_t) \leq \sum_{t=0}^{T-1} b_t H_t \Pr_{\widehat{\pi}}(X_t \in E_t). \tag{36}$$

The first events are disjoint. Any independent bound on the entire cash-cost difference may be intersected with this bound. To apply it to an optimization class, the action agreement and bad sets must hold uniformly for every policy in that class.

PROOF: On D_t, the histories, states and costs before t coincide. The absolute difference of the remaining discounted costs is at most $b_t H_t$. On the event of no disagreement the costs coincide throughout. Summing over the disjoint first-disagreement events and taking expectations proves the first inequality. Up to the first disagreement, the common state is the state of the reference policy driven by the same innovations. A first disagreement at date t requires this state to lie in E_t, proving the second inequality. *Q.E.D.*

This proposition is a cost-transfer identity, not a substitute for constructing longer-horizon stable sublevels. In particular, changing a terminal readout changes labels upstream. Independently small optimization gaps at different fitted label problems do not automatically establish the uniform action-agreement premise. In the executed two-date experiment the last-date actor depends only on the common terminal payoff; Theorem 10 therefore needs the first-date switch problems alone.

Y.3. *Confidence and economic interpretation*

The original cost event concerns a finite list of frozen attained policies. On that event all deterministic stability enlargements are valid simultaneously, including for a coefficient vector selected after seeing the records but still belonging to a certified fixed-label sublevel. The probability statement does not change because the optimizer class is uncountable. A new unqualified model or newly fitted label problem is not covered by that argument.

A failed action-stability certificate is a numerical statement about the available lower bound, not a witness of policy variation. Conversely, policy invariance of an exact optimization class does not assert that an arbitrary practical optimizer reaches that class. The deposited feasible iterate and dual gap describe what the construction attained; the switch lower bounds describe what every member of a stated sublevel must do. These are distinct quantifiers, and neither is inferred from a collection of fitting seeds.

For the two calibrated $n = 16$ controllers let $[l, u]$ be the positive interval for ReLU cost minus Bernstein cost. Let κ be the additional per-deployment charge for Bernstein, after all actual installation and computation charges have been converted into the same normalized cost units. The corresponding net difference is $[l - \kappa, u - \kappa]$. Bernstein is strictly less costly for every $\kappa < l$; ReLU is strictly less costly for every $\kappa > u$; neither strict ordering is certified in the intermediate region. This is an exact conditional price region, not a monetary calibration. It remains valid for the fully action-stable optimizer classes, and it does not use construction times with different service boundaries as interchangeable monetary prices.

APPENDIX Z: COMPLETE FACTORIAL AND PRECISION RECORDS

The following tables report both estimators, all marginal costs, paired contrasts and predeclared interactions. A paired-width target is not a marginal-cost target. The two estimators share the same underlying rows and target the same expected costs; no independent sample size is gained by computing both. The interval width is decomposed into the two endpoint sampling-variance allowances, the bounded-support term and the mean numerical enclosure width. Arithmetic, continuous-bin

width, and any acquisition branching enter the latter through actual endpoint propagation. The implementation does not claim a uniquely identifiable additive attribution of that combined numerical width to each intermediate operation.

TABLE Z.I

OBSERVED WIDTH COMPONENTS FOR THE COMPLETE FROZEN FAMILY

Estimator	Type	Count	Max width	Support	Variance	Numerical
Cash	Marginal	9	0.006901	0.000505	0.006397	1.40e-11
Cash	Pair	36	0.002443	0.001009	0.001434	2.79e-11
Cash	Interaction	2	0.002913	0.002018	0.000895	5.52e-11
Conditional	Marginal	9	0.006901	0.000505	0.006397	1.35e-11
Conditional	Pair	36	0.002441	0.001009	0.001433	2.70e-11
Conditional	Interaction	2	0.002912	0.002018	0.000895	5.34e-11

Maxima are taken within each row and need not occur in the same estimand. Support and variance are the two-sided finite-sample allowances. Numerical is the mean endpoint-enclosure width, inclusive of propagated arithmetic and bin effects. The predeclared paired width target is 1/256; marginal intervals are not governed by that paired target. Components are not counted twice.

TABLE Z.III: Every original-law cost estimand in the R80 simultaneous family

Estimator	Estimand	Interval	Width target
Cash	CB16	[0.598996, 0.604724]	–
Cash	CB8	[0.601755, 0.607492]	–
Cash	CR16	[0.600489, 0.606198]	–
Cash	CR8	[0.605889, 0.611618]	–
Cash	LB16	[0.682148, 0.688581]	–
Cash	LB8	[0.683269, 0.689716]	–
Cash	LR16	[0.606271, 0.611918]	–
Cash	LR8	[0.606970, 0.612648]	–
Cash	Z	[0.772792, 0.779694]	–
Cash	CB8 minus CB16	[0.002241, 0.003286]	Pass

Estimator	Estimand	Interval	Width target
Cash	CB8 minus CR16	[0.000748, 0.001812]	Pass
Cash	CR16 minus CB16	[0.000955, 0.002012]	Pass
Cash	CR8 minus CB16	[0.006337, 0.007450]	Pass
Cash	CR8 minus CB8	[0.003572, 0.004689]	Pass
Cash	CR8 minus CR16	[0.004855, 0.005965]	Pass
Cash	LB16 minus CB16	[0.082602, 0.084406]	Pass
Cash	LB16 minus CB8	[0.079844, 0.081638]	Pass
Cash	LB16 minus CR16	[0.081108, 0.082933]	Pass
Cash	LB16 minus CR8	[0.075693, 0.077528]	Pass
Cash	LB8 minus CB16	[0.083723, 0.085541]	Pass
Cash	LB8 minus CB8	[0.080964, 0.082773]	Pass
Cash	LB8 minus CR16	[0.082229, 0.084068]	Pass
Cash	LB8 minus CR8	[0.076814, 0.078663]	Pass
Cash	LB8 minus LB16	[0.000613, 0.001643]	Pass
Cash	LB8 minus LR16	[0.076429, 0.078368]	Pass
Cash	LR16 minus CB16	[0.006635, 0.007834]	Pass
Cash	LR16 minus CB8	[0.003870, 0.005072]	Pass
Cash	LR16 minus CR16	[0.005159, 0.006342]	Pass
Cash	LR16 minus CR8	[-0.000271, 0.000952]	Pass
Cash	LR16 minus LB16	[-0.077233, -0.075307]	Pass
Cash	LR8 minus CB16	[0.007365, 0.008532]	Pass
Cash	LR8 minus CB8	[0.004602, 0.005769]	Pass
Cash	LR8 minus CR16	[0.005889, 0.007041]	Pass
Cash	LR8 minus CR8	[0.000457, 0.001653]	Pass
Cash	LR8 minus LB16	[-0.076498, -0.074612]	Pass
Cash	LR8 minus LB8	[-0.077633, -0.075734]	Pass
Cash	LR8 minus LR16	[0.000186, 0.001243]	Pass
Cash	Z minus CB16	[0.173223, 0.175544]	Pass
Cash	Z minus CB8	[0.170464, 0.172776]	Pass
Cash	Z minus CR16	[0.171729, 0.174071]	Pass
Cash	Z minus CR8	[0.166315, 0.168664]	Pass
Cash	Z minus LB16	[0.090114, 0.091644]	Pass
Cash	Z minus LB8	[0.088993, 0.090509]	Pass

Estimator	Estimand	Interval	Width target
Cash	Z minus LR16	[0.165927, 0.168370]	Pass
Cash	Z minus LR8	[0.165232, 0.167636]	Pass
Cash	Interaction 16	[-0.079210, -0.076297]	Pass
Cash	Interaction 8	[-0.082255, -0.079372]	Pass
Conditional	CB16	[0.598987, 0.604714]	–
Conditional	CB8	[0.601746, 0.607483]	–
Conditional	CR16	[0.600481, 0.606189]	–
Conditional	CR8	[0.605880, 0.611609]	–
Conditional	LB16	[0.682138, 0.688570]	–
Conditional	LB8	[0.683259, 0.689705]	–
Conditional	LR16	[0.606262, 0.611908]	–
Conditional	LR8	[0.606961, 0.612638]	–
Conditional	Z	[0.772780, 0.779682]	–
Conditional	CB8 minus CB16	[0.002241, 0.003286]	Pass
Conditional	CB8 minus CR16	[0.000748, 0.001812]	Pass
Conditional	CR16 minus CB16	[0.000955, 0.002013]	Pass
Conditional	CR8 minus CB16	[0.006337, 0.007450]	Pass
Conditional	CR8 minus CB8	[0.003571, 0.004688]	Pass
Conditional	CR8 minus CR16	[0.004854, 0.005965]	Pass
Conditional	LB16 minus CB16	[0.082601, 0.084405]	Pass
Conditional	LB16 minus CB8	[0.079843, 0.081637]	Pass
Conditional	LB16 minus CR16	[0.081107, 0.082932]	Pass
Conditional	LB16 minus CR8	[0.075693, 0.077527]	Pass
Conditional	LB8 minus CB16	[0.083722, 0.085540]	Pass
Conditional	LB8 minus CB8	[0.080964, 0.082772]	Pass
Conditional	LB8 minus CR16	[0.082228, 0.084067]	Pass
Conditional	LB8 minus CR8	[0.076814, 0.078662]	Pass
Conditional	LB8 minus LB16	[0.000613, 0.001643]	Pass
Conditional	LB8 minus LR16	[0.076428, 0.078367]	Pass
Conditional	LR16 minus CB16	[0.006635, 0.007834]	Pass
Conditional	LR16 minus CB8	[0.003869, 0.005072]	Pass
Conditional	LR16 minus CR16	[0.005159, 0.006341]	Pass
Conditional	LR16 minus CR8	[-0.000271, 0.000952]	Pass

Estimator	Estimand	Interval	Width target
Conditional	LR16 minus LB16	[-0.077232, -0.075307]	Pass
Conditional	LR8 minus CB16	[0.007365, 0.008532]	Pass
Conditional	LR8 minus CB8	[0.004602, 0.005768]	Pass
Conditional	LR8 minus CR16	[0.005889, 0.007041]	Pass
Conditional	LR8 minus CR8	[0.000457, 0.001653]	Pass
Conditional	LR8 minus LB16	[-0.076497, -0.074612]	Pass
Conditional	LR8 minus LB8	[-0.077633, -0.075733]	Pass
Conditional	LR8 minus LR16	[0.000186, 0.001243]	Pass
Conditional	Z minus CB16	[0.173220, 0.175540]	Pass
Conditional	Z minus CB8	[0.170461, 0.172772]	Pass
Conditional	Z minus CR16	[0.171726, 0.174067]	Pass
Conditional	Z minus CR8	[0.166313, 0.168660]	Pass
Conditional	Z minus LB16	[0.090113, 0.091641]	Pass
Conditional	Z minus LB8	[0.088991, 0.090506]	Pass
Conditional	Z minus LR16	[0.165925, 0.168367]	Pass
Conditional	Z minus LR8	[0.165230, 0.167633]	Pass
Conditional	Interaction 16	[-0.079209, -0.076297]	Pass
Conditional	Interaction 8	[-0.082254, -0.079371]	Pass

Codes: Z is zero control; L and C denote legacy and calibrated qualification, R and B denote ReLU and Bernstein, and the final number is the acquisition resolution. Display endpoints are rounded outward. Full exact endpoints, moments, equivalence flags and both numerical implementations remain in the machine-readable records.

APPENDIX: REFERENCES

AGRAWAL, AKSHAY, SHANE BARRATT, STEPHEN BOYD, AND BARTOLOMEO STELLATO (2020): "Learning Convex Optimization Control Policies," in *Proceedings of the 2nd Conference on Learning for Dynamics and Control*, 120, 361–373. [67]

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TABLE Z.II

CONSTRUCTION OPERATION COUNTS AND RECORDED SERVICE BOUNDARIES

Actor	Action values	Feature values	Seconds	Label ablation
CB16	33792	152064	6.269	No
CB8	8448	38016	1.547	No
CR16	33792	152064	2.699	No
CR8	8448	38016	0.653	No
LB16	33792	152064	10.617	Yes
LB8	8448	38016	3.174	Yes
LR16	33792	152064	4.793	Yes
LR8	8448	38016	1.174	Yes

Legacy clocks include a matched fixed-label loss ablation; calibrated clocks do not. Hence this table does not compare identical end-to-end timing services. All reported counts and clocks come from their original construction records. The shared cost-inference service, independent replay, barrier verification and publication work have separate clocks and are not assigned zero cost.

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SAMBHARYA, RAJIV, GEORGINA HALL, BRANDON AMOS, AND BARTOLOMEO STELLATO (2023): "End-to-End Learning to Warm-Start for Real-Time Quadratic Optimization," in *Proceedings of the 5th Annual Learning for Dynamics and Control Conference*, 211, 220–234. [71]